TOPOLOGY

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Contents

The purpose of this document is to compile common notions in basic topology.

1. Basic topology

1.1. Topological spaces.

Definition 1.1.1 (Topology)**.** Let X be a set. A *topology on X* is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i\in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i\in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \cdots \cap U_n$ is in $\mathcal{T}$. Equivalently, for any $U_1, U_2 \in \mathcal{T}$, the intersection $U_1 \cap U_2$ is in $\mathcal{T}$

If $\mathcal{T}$ is a topology on X, the pair $(X, \mathcal{T})$ is called a *topological space*. Members of $\mathcal{T}$ are called *open sets*.

A subset $C \subseteq X$ is *closed* (also called a *closed subset*) if its complement $X \setminus C$ is an open set in $\mathcal{T}$

One very often refers to X as a topological spcae, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as $\mathrm{Top}(X)$, $\mathbf{Top}(X)$, or $\mathsf{Top}(X)$.

Definition 1.1.2. Let X be a topological space. A subset $U \subseteq X$ is a *clopen subset* if U is both an open set and a closed set in the topology of X.

1.1.1. *Generating topologies from subsets.*

Definition 1.1.3. Let X be a set.

1. Let τ_1, τ_2 be topologies on (Definition 1.1.1) X. Say that τ_1 is *coarser than* (equivalently, *smaller than*) τ_2 if $\tau_1 \subseteq \tau_2$, and that τ_1 is *finer than* (equivalently, *larger than*) τ_2 if $\tau_2 \subseteq \tau_1$. This means that every subset of X that is open (Definition 1.1.1) with respect to τ_1 is also open with respect to τ_2.
   These relations are denoted by $\tau_1 \preceq \tau_2$ for "τ_1 coarser than τ_2" and $\tau_1 \succeq \tau_2$ for "τ_1 finer than τ_2"; their strict versions are $\tau_1 \prec \tau_2$ and $\tau_1 \succ \tau_2$, meaning proper inclusion.
2. Let $\mathcal{C}$ be some family of topologies on X. A topology $\tau \in \mathcal{C}$ is the *coarsest* (or *smallest*) element of $\mathcal{C}$ if for every $\sigma \in \mathcal{C}$ one has $\tau \subseteq \sigma$ (equivalently, $\tau \preceq \sigma$ for all $\sigma \in \mathcal{C}$). Dually, $\tau \in \mathcal{C}$ is the *finest* (or *largest*) element of $\mathcal{C}$ if for every $\sigma \in \mathcal{C}$ one has $\sigma \subseteq \tau$ (equivalently, $\sigma \preceq \tau$ for all $\sigma \in \mathcal{C}$).

Proposition 1.1.4. Let X be a set and let $f : (X, \tau_X) \to (Y, \tau_Y)$ be a function between topological spaces (Definition 1.1.1).

- If τ'_X is a topology (Definition 1.1.1) on X finer (Definition 1.1.3) than τ_X and f is continuous (Definition 1.3.1), then $f : (X, \tau'_X) \to (Y, \tau_Y)$ is also continuous.
- If τ'_Y is a topology on Y coarser than τ_Y and f is continuous, then $f : (X, \tau_X) \to (Y, \tau'_Y)$ is also continuous.

Proposition 1.1.5. Let X be a set and $\mathcal{S}$ a collection of subsets of X. The topology (Definition 1.1.1) $\tau(\mathcal{S})$ generated by (Definition 1.1.6) $\mathcal{S}$ is the coarsest topology (Definition 1.1.3) on X such that every element of $\mathcal{S}$ is an open (Definition 1.1.1) set.

Definition 1.1.6. Let X be a set and $\mathcal{S}$ a collection of subsets of X. The *topology on X generated by $\mathcal{S}$*, often denoted $\tau(\mathcal{S})$, is the intersection of all topologies on X that contain $\mathcal{S}$. It is the unique coarsest topology (Definition 1.1.3) on X in which every element of $\mathcal{S}$ is an open set.

The collection $\mathcal{S}$ is called a *subbasis for the topology $\tau(\mathcal{S})$*. Alternatively, one might generally call a family $\mathcal{S} \subseteq \mathcal{P}(X)$ a *subbasis on X* if $\bigcup_{S \in \mathcal{S}} S = X$ and we would consider $\mathcal{S}$ the subbasis for the topology $\tau(\mathcal{S})$.

Proposition 1.1.7. Let X be a set and $\mathcal{S}$ a subbasis (Definition 1.1.6) for a topology (Definition 1.1.1) τ. A subset $U \subseteq X$ belongs to τ if and only if U is the union of finite intersections of elements of $\mathcal{S}$. That is, the collection of all finite intersections of elements of $\mathcal{S}$ forms a basis (Definition 1.1.17) for τ.

Proposition 1.1.8. Let X and Y be topological spaces (Definition 1.1.1), and let $\mathcal{S}$ be a subbasis (Definition 1.1.6) for the topology (Definition 1.1.1) on Y. A function $f : X \to Y$ is continuous (Definition 1.3.1) if and only if $f^{-1}(V)$ is an open (Definition 1.1.1) set in X for every $V \in \mathcal{S}$.

1.1.2. *Initial and final topologies.*

Definition 1.1.9 (Initial topology)**.** (♠ TODO: AI generated, verify) Let S be a set and let $\{X_i\}_{i\in I}$ be a family of topological spaces (Definition 1.1.1). Suppose that for each $i \in I$, a function

$$f_i : S \to X_i$$

is given. The *initial topology on S with respect to the family $\{f_i\}_{i\in I}$* is the coarsest topology (Definition 1.1.3) on S such that every map $f_i : S \to X_i$ is continuous (Definition 1.3.1).

Specifically, the initial topology is generated by the subbasis consisting of all sets of the form

$$f_i^{-1}(U),$$

where $i \in I$ and U ranges over all open subsets of X_i. Equivalently, a filter $\mathcal{F}$ on S converges to $x \in S$ in the initial topology if and only if the pushed-forward filters $(f_i)_*(\mathcal{F})$ converge to $f_i(x)$ in X_i for every $i \in I$.

The resulting topological space is called the *initial topological space induced by $\{f_i\}_{i\in I}$*. When no confusion can arise, we simply say "the initial topology" and refer to the family $\{f_i\}_{i\in I}$ as the *structure maps* or *defining family*.

The initial topology is the limit in the category of topological spaces of the diagram consisting of the codomain spaces $\{X_i\}_{i\in I}$ together with the structure maps. In particular, it satisfies the following universal property: a map $g : Y \to S$ from any topological space Y is continuous if and only if each composite $f_i \circ g : Y \to X_i$ is continuous for all $i \in I$.

Definition 1.1.10 (Final topology)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $\{f_i : X_i \to X\}_{i\in I}$ be a family of maps from topological spaces to a set X. The *final topology on X with respect to the family $\{f_i\}_{i\in I}$* is the finest topology on X such that every map $f_i : X_i \to X$ is continuous.

Specifically, a subset $U \subseteq X$ is open in the final topology if and only if $f_i^{-1}(U)$ is open in X_i for every $i \in I$. Equivalently, a net (x_α) in X converges to $x \in X$ in the final topology if and only if there exists some index $j \in I$, a point $y \in X_j$, and a net (y_β) in X_j converging to y such that $f_j(y_\beta) = x_{\alpha(\beta)}$ for all β and $f_j(y) = x$.

The resulting topological space is called the *final topological space induced by $\{f_i\}_{i\in I}$*. When no confusion can arise, we simply say "the final topology" and refer to the family $\{f_i\}_{i\in I}$ as the *structure maps* or *defining family*.

The final topology is the colimit in the category of topological spaces of the diagram consisting of the domain spaces $\{X_i\}_{i\in I}$ together with the structure maps. In particular, it satisfies the following universal property: a map $g : X \to Y$ to any topological space Y is continuous if and only if each composite $g \circ f_i : X_i \to Y$ is continuous for all $i \in I$.

Proposition 1.1.11 (Universal property of initial and final topologies)**.** (♠ TODO: AI generated, verify)

Let S be a set, $\{X_i\}_{i\in I}$ a family of topological spaces, and $\{f_i : S \to X_i\}_{i\in I}$ a family of functions. If S is equipped with the initial topology with respect to this family, then for any topological space Y, a function $g : Y \to S$ is continuous if and only if each composite $f_i \circ g : Y \to X_i$ is continuous for all $i \in I$.

Dually, let X be a topological space and $\{f_i : X_i \to X\}_{i\in I}$ a family of functions. If X carries the final topology with respect to this family, then for any topological space Y, a function $g : X \to Y$ is continuous if and only if each composite $g \circ f_i : X_i \to Y$ is continuous for all $i \in I$.

1.1.3. *Quotient and disjoint union topologies.*

Definition 1.1.12 (Quotient topology)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $\sim$ be an equivalence relation on the underlying set of X. Write $\pi : X \to X/\sim$ for the canonical quotient map that sends each point to its equivalence class. The *quotient topology on $X/\sim$* is the unique topology on the quotient set $X/\sim$ characterized by any of the following equivalent conditions:

1. A subset $U \subseteq X/\sim$ is open if and only if $\pi^{-1}(U)$ is open in X.
2. The quotient topology is the final topology (Definition 1.1.10) with respect to the single map $\pi : X \to X/\sim$.
3. A function $f : X/\sim \to Y$ from the quotient space to any topological space Y is continuous if and only if the composition $f \circ \pi : X \to Y$ is continuous.

The resulting topological space $(X/\sim, \mathcal{T}_{\text{quot}})$ is called the *quotient space of X by $\sim$*.

More generally, let X be a topological space and let $f : X \to Y$ be any surjective map to a set Y. The *quotient topology on Y induced by f* is the final topology with respect to f, i.e., a subset $U \subseteq Y$ is open if and only if $f^{-1}(U)$ is open in X.

Definition 1.1.13 (Disjoint union topology)**.** Let $\{X_i\}_{i\in I}$ be an arbitrary small collection of topological spaces (Definition 1.1.1). The *disjoint union space* (or *topological sum*), denoted by $\coprod_{i\in I} X_i$, is the set-theoretic disjoint union $\bigcup_{i\in I}(X_i \times \{i\})$ endowed with the *disjoint union topology*, which is the finest topology (Definition 1.1.3) such that each canonical injection

$$\iota_j : X_j \hookrightarrow \coprod_{i\in I} X_i, \quad x \mapsto (x, j)$$

is continuous (Definition 1.3.1).

Specifically, a subset $U \subseteq \coprod_{i\in I} X_i$ is open (Definition 1.1.1) if and only if $\iota_i^{-1}(U)$ is open in X_i for every $i \in I$. The disjoint union space is the coproduct in the category of topological spaces.

Proposition 1.1.14 (Universal property of quotient topology)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $\sim$ be an equivalence relation on the underlying set. Write $\pi : X \to X/\sim$ for the canonical quotient map, where $X/\sim$ carries the quotient topology. Then for any topological space Y, a function $f : X/\sim \to Y$ is continuous if and only if the composition $f \circ \pi : X \to Y$ is continuous.

More generally, let X be a topological space, Y a set, and $f : X \to Y$ any surjective map. Endow Y with the quotient topology induced by f. Then for any topological space Z, a function $g : Y \to Z$ is continuous if and only if $g \circ f : X \to Z$ is continuous.

Proposition 1.1.15 (Universal property of disjoint union topology)**.** (♠ TODO: AI generated, verify) Let $\{X_i\}_{i \in I}$ be a family of topological spaces and let $\coprod_{i \in I} X_i$ be their disjoint union with the disjoint union topology. Then for any topological space Y, a function $f : \coprod_{i \in I} X_i \to Y$ is continuous if and only if each composite $f \circ \iota_j : X_j \to Y$ is continuous for all $j \in I$, where $\iota_j : X_j \hookrightarrow \coprod_{i \in I} X_i$ denotes the canonical injection.

Equivalently, a function $f : Y \to \coprod_{i \in I} X_i$ is continuous if and only if each component map $p_j \circ f : Y \to X_j$ is continuous for all $j \in I$, where p_j denotes the projection onto the j-th coordinate (defined on the image of ι_j).

1.1.4. *Basis of a topological space.*

Definition 1.1.16. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1) and let $x \in X$.

- An *open neighborhood of x* is any open set $U \in \mathcal{T}$ such that $x \in U$.
- A *neighborhood of x* is a set $N \subseteq X$ for which there exists an open neighborhood $U \in \mathcal{T}$ of x such that $U \subseteq N$.
- A *neighborhood basis* (or *local base*) at x is a nonempty collection $\mathcal{B}_x$ of neighborhoods of x such that for every neighborhood N of x, there exists $B \in \mathcal{B}_x$ with $B \subseteq N$.
  The elements of $\mathcal{B}_x$ are said to *form a base of neighborhoods* at x.

Definition 1.1.17. Let X be a set and let $\mathcal{B}$ be a collection of subsets of X. The collection $\mathcal{B}$ is called a *basis* (or *base*) for a topology (Definition 1.1.1) on X if the following two conditions hold:

1. For every $x \in X$, there exists at least one $B \in \mathcal{B}$ such that $x \in B$.
2. For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Given such a collection $\mathcal{B}$, the collection $\mathcal{T}$ of all unions of elements of $\mathcal{B}$ defines a topology on X, and it coincides with $\mathcal{T}_{\mathcal{B}}$, the topology generated by $\mathcal{B}$ (Definition 1.1.6). In other words,

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \text{for every } x \in U, \text{ there exists } B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$

Definition 1.1.18. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1).

- The space $(X, \mathcal{T})$ is said to be *first countable* if every point $x \in X$ has a countable (Definition A.0.1) neighborhood basis (Definition 1.1.16), i.e., there exists a countable collection $\{U_n\}_{n \in \mathbb{N}}$ of open neighborhoods of x such that for any open neighborhood U of x, there exists $n \in \mathbb{N}$ with $U_n \subseteq U$.

- The space $(X, \mathcal{T})$ is said to be *second countable* if there exists a countable basis (Definition 1.1.17) $\mathcal{B} \subseteq \mathcal{T}$ for the topology $\mathcal{T}$, i.e., every open set $U \in \mathcal{T}$ can be written as a union of elements of $\mathcal{B}$.

1.1.5. *Function space topologies.*

Definition 1.1.19. Let X and Y be topological spaces (Definition 1.1.1).

1. Write $\mathrm{Open}(Y)$ for the topology (Definition 1.1.1) of Y and $\mathrm{Comp}(X)$ for the family of compact subsets of X. For $K \in \mathrm{Comp}(X)$ and $U \in \mathrm{Open}(Y)$, define the subset
$$\langle K, U\rangle \coloneqq \{ f \in C(X,Y) \mid f(K) \subseteq U \} \subseteq C(X,Y).$$
where $C(X,Y)$ (Definition 1.3.1) is the set of all continuous maps (Definition 1.3.1) $X \to Y$. The collection of all such subsets is denoted by $\mathcal{S}_{\mathrm{co}}(X,Y)$.
2. The *compact-open topology on $C(X,Y)$* (Definition 1.3.1) is the topology (Definition 1.1.1) generated by (Definition 1.1.6) the subbasis (Definition 1.1.6) $\mathcal{S}_{\mathrm{co}}(X,Y)$. The resulting topological space is called the *function space from X to Y* and is commonly denoted by notations such as $\mathrm{Map}(X,Y)$, Y^X, $\mathrm{Fun}(X,Y)$, or $\mathrm{F}(X,Y)$. As a set, it equals $C(X,Y)$.
3. Let (X, x_0) and (Y, y_0) be based topological spaces (Definition 1.3.3). The *based function space from (X, x_0) to (Y, y_0)* is the subspace of $\mathrm{Map}(X,Y)$ consisting of all based maps f with $f(x_0) = y_0$, equivalently $C_*((X,x_0),(Y,y_0))$ endowed with the subspace topology inherited from the compact-open function space $\mathrm{Map}(X,Y)$. It itself is a based topological space whose base point is given by the constant map $X \to Y$ with value y_0. Common notations for the based function space include those which include a star or bullet such as $\mathrm{Map}_*((X,x_0),(Y,y_0))$, $\mathrm{Map}_*(X,Y)$, or Y^X_* to emphasize that the spaces involved are pointed, or those which omit a star or bullet, such as $\mathrm{Map}(X,Y)$, and Y^X.

Definition 1.1.20 (Weak topology on a vector space)**.** (♠ TODO: AI generated, verify)

Let V be a vector space over a field k, and let $\langle V, V^*\rangle : V \times V^* \to k$ denote the canonical bilinear pairing between V and its algebraic dual V^*. The *weak topology on V* is the initial topology (Definition 1.1.9) with respect to the family of all linear functionals $\{f : V \to k\}_{f \in V^*}$, where each copy of k carries its standard topology.

Equivalently, a net $(x_\alpha)_{\alpha \in D}$ in V converges weakly to $x \in V$ if and only if $f(x_\alpha) \to f(x)$ in k for every $f \in V^*$. A subset $C \subseteq V$ is called *weakly closed* if it is closed in the weak topology.

When V carries an additional norm (or more generally a locally convex topology), the *continuous dual space* V' of continuous linear functionals may be smaller than the full algebraic dual V^*. In this case, one defines the *weak topology relative to the pairing $\langle V, V'\rangle$* as the initial topology with respect to $\{f : V \to k\}_{f \in V'}$. The weak topology is always coarser than the original norm (or locally convex) topology on V, and they coincide if and only if V is finite-dimensional.

Definition 1.1.21 (Weak* topology on a dual space)**.** (♠ TODO: AI generated, verify)

Let V be a vector space over a field k, and let V^* denote its algebraic dual. Fix a pairing $\langle V, V^* \rangle : V \times V^* \to k$ given by evaluation: $\langle v, f \rangle = f(v)$ for $v \in V$ and $f \in V^*$. The *weak* topology on V^** is the initial topology (Definition 1.1.9) with respect to the family of evaluation maps $\{\mathrm{ev}_v : V^* \to k\}_{v \in V}$, where each copy of k carries its standard topology.

Equivalently, a net $(f_\alpha)_{\alpha \in D}$ in V^* converges to $f \in V^*$ in the weak* topology if and only if $f_\alpha(v) \to f(v)$ in k for every $v \in V$. A subset $C \subseteq V^*$ is called *weak*-closed* if it is closed in the weak* topology.

When V carries a norm, we write V' for its continuous dual and V^{**} for the bidual. The weak* topology on V^* is always coarser than the original norm topology (the topology induced by operator norms), and they coincide if and only if V is finite-dimensional.

By the Banach–Alaoglu theorem, when $k = \mathbb{R}$ or $\mathbb{C}$ and V is a normed vector space, the closed unit ball in V^* is compact in the weak* topology.

Proposition 1.1.22 (Weak topology is the initial topology with respect to all continuous linear functionals)**.** (♠ TODO: AI generated, verify) Let V be a topological vector space over $\mathbb{R}$ or $\mathbb{C}$, and let V' denote its continuous dual space. The weak topology on V (induced by the pairing $\langle V, V' \rangle$) coincides with the initial topology on V with respect to the family of all linear functionals $\{f : V \to k\}_{f \in V'}$, where each copy of k carries its standard Euclidean topology.

In particular, a net $(x_\alpha)_{\alpha \in D}$ in V converges weakly to $x \in V$ if and only if $f(x_\alpha) \to f(x)$ for every $f \in V'$.

Proposition 1.1.23 (Weak* topology is the coarsest making evaluations continuous)**.** Let V be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$, and let V' denote its continuous dual. The weak* topology on V' is the coarsest (i.e., smallest) topology such that every evaluation map $\mathrm{ev}_v : V' \to k$ defined by $\mathrm{ev}_v(f) = f(v)$ for $v \in V$ is continuous, where each copy of k carries its standard Euclidean topology.

Equivalently, the weak* topology on V' coincides with the initial topology with respect to the family of evaluation maps $\{\mathrm{ev}_v : V' \to k\}_{v \in V}$. In particular, a net $(f_\alpha)_{\alpha \in D}$ in V' converges weak* to $f \in V'$ if and only if $f_\alpha(v) \to f(v)$ for every $v \in V$.

1.1.6. *Separation axioms.*

Definition 1.1.24. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1) and let $A \subseteq X$.

- The set A is called a *G_δ set* if there exists a countable collection of open sets $\{U_n\}_{n \in \mathbb{N}} \subseteq \mathcal{T}$ such that
$$A = \bigcap_{n \in \mathbb{N}} U_n .$$
- The set A is called an *F_σ set* if there exists a countable collection of closed sets $\{F_n\}_{n \in \mathbb{N}}$ such that
$$A = \bigcup_{n \in \mathbb{N}} F_n .$$

Definition 1.1.25 (Separation axioms)**.** Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1).

- $(X, \mathcal{T})$ is *T_0* (*Kolmogorov*) if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is *T_1* (*Fréchet*) if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is *T_2* or *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is *T_3* (regular Hausdorff) if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \to [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is *$T_{3\frac{1}{2}}$* (completely regular Hausdorff) if it is T_1 and completely regular.
- $(X, \mathcal{T})$ is *normal* if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is *T_4* (normal Hausdorff) if it is T_1 and normal.
- $(X, \mathcal{T})$ is *T_5* (completely normal Hausdorff) if it is T1 and completely normal.
- $(X, \mathcal{T})$ is *perfectly normal* if every closed set is a $G\delta$ (Definition 1.1.24) (countable (Definition A.0.1) intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is *T_6* (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

1.1.7. *Subspaces of a topological space.*

Definition 1.1.26 (Subspace topology and subspace)**.** Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a subset. The topology induced on Y by $\mathcal{T}$, called the *subspace topology*, is defined as

$$\mathcal{T}_Y := \{U \cap Y : U \in \mathcal{T}\}.$$

The pair $(Y, \mathcal{T}_Y)$ is called a *subspace of* $(X, \mathcal{T})$.

Definition 1.1.27. Let $f : X \to Y$ be a continuous map (Definition 1.3.1) of topological spaces (Definition 1.1.1).

The image $f(X)$ of f is, by default, regarded as a topological space by equipping $f(X)$ with its subspace topology (Definition 1.1.26) in Y.

Convention 1.1.28. Given a topological space $(X, \mathcal{T})$ and a subset S of X, one almost always considers S as a subspace of X, i.e. considers S as a topological space equipped with the subspace topology.

Definition 1.1.29 (Discrete topological space)**.** Let X be a set. The *discrete topology on X* is the topology $\mathcal{T}_{\text{disc}}$ defined by

$$\mathcal{T}_{\text{disc}} := \{U : U \subseteq X\},$$

i.e., every subset of X is open. The pair $(X, \mathcal{T}_{\text{disc}})$ is called a *discrete topological space* or even a *discrete set*.

Definition 1.1.30 (Indiscrete (or trivial) topological space)**.** Let X be a set. The *indiscrete topology* (also called the *trivial topology*) on X is the topology $\mathcal{T}_{\text{indisc}}$, also denoted by notations such as $\mathcal{T}_{\text{triv}}$, defined by

$$\mathcal{T}_{\text{indisc}} := \{\emptyset, X\},$$

i.e., only the empty set and the entire set are open. The pair $(X, \mathcal{T}_{\text{indisc}})$ is called an *indiscrete topological space*.

Definition 1.1.31 (Closure of a subset)**.** Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$ a subset. The *closure of A in X*, denoted by $\overline{A}$, is defined as the intersection of all closed sets containing A, i.e.,

$$\overline{A} := \bigcap\{C \subseteq X : C \text{ is closed and } A \subseteq C\}.$$

Equivalently, $\overline{A}$ is the smallest closed set containing A.

1.2. **Properties of topological spaces.**

Definition 1.2.1 (Dense subset)**.** Let $(X, \mathcal{T})$ be a topological space. A subset $D \subseteq X$ is called *dense in X* if the closure of D equals X, i.e.,

$$\overline{D} = X.$$

Equivalently, D is dense in X if every non-empty open set $U \in \mathcal{T}$ intersects D, that is,

$$\forall U \in \mathcal{T},\ U \neq \emptyset \implies U \cap D \neq \emptyset.$$

1.2.1. *Nets and filters.*

Definition 1.2.2 (Directed set)**.** (♠ TODO: AI generated, verify)

A nonempty set D equipped with a binary relation $\leq$ is called a *directed set* (or *directed preorder*) if the following conditions hold:

1. **Reflexivity:** For every $\alpha \in D$, we have $\alpha \leq \alpha$.
2. **Transitivity:** For every $\alpha, \beta, \gamma \in D$, if $\alpha \leq \beta$ and $\beta \leq \gamma$, then $\alpha \leq \gamma$.
3. **Upper bound property:** For every pair $\alpha, \beta \in D$, there exists $\gamma \in D$ such that $\alpha \leq \gamma$ and $\beta \leq \gamma$.

A subset $E \subseteq D$ is called a *cofinal subset* of D if for every $\alpha \in D$, there exists $\beta \in E$ such that $\alpha \leq \beta$.

Definition 1.2.3 (Net in a topological space)**.** (♠ TODO: AI generated, verify)

Let D be a directed set (Definition 1.2.2). A *net in a topological space X indexed by D* is a function from D to X, i.e., a family $(x_\alpha)_{\alpha \in D}$ of points of X. We often write simply "a net" when the indexing set and ambient space are clear from context.

Nets generalize sequences: a sequence is exactly a net indexed by $\mathbb{N}$ with its usual ordering. Unlike sequences, nets can have uncountable or otherwise non-well-ordered directed sets as indices, which allows them to capture convergence in arbitrary topological spaces.

Definition 1.2.4 (Convergence of a net)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $(D, \leq)$ be a directed set (Definition 1.2.2). Consider a net $(x_\alpha)_{\alpha \in D}$ in X.

We say that the net $(x_\alpha)_{\alpha \in D}$ *converges* to a point $x \in X$ (written $x_\alpha \to x$ or $\lim_{\alpha \in D} x_\alpha = x$) if for every open neighborhood U of x, there exists $\alpha_0 \in D$ such that for all $\alpha \geq \alpha_0$, we have $x_\alpha \in U$.

The point x is called a *limit* of the net $(x_\alpha)_{\alpha \in D}$. A net may have multiple limits if X is not Hausdorff; in a Hausdorff space, every convergent net has at most one limit.

Nets are significant because they provide a complete characterization of topological spaces:

1. A subset $C \subseteq X$ is closed if and only if it contains the limits of all convergent nets in C.
2. A function $f : X \to Y$ between topological spaces is continuous if and only if for every net $(x_\alpha)_{\alpha \in D}$ in X converging to x, the net $(f(x_\alpha))_{\alpha \in D}$ converges to $f(x)$ in Y.

Definition 1.2.5 (Filter on a set)**.** (♠ TODO: AI generated, verify)

Let S be a nonempty set. A nonempty collection $\mathcal{F}$ of subsets of S is called a *filter on S* if it satisfies the following conditions:

1. The empty set is not in $\mathcal{F}$, i.e., $\emptyset \notin \mathcal{F}$.
2. **Upward closure:** If $A \in \mathcal{F}$ and $A \subseteq B \subseteq S$, then $B \in \mathcal{F}$.
3. **Finite intersection property:** If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

The elements of a filter are called its *members*. A filter $\mathcal{F}$ is said to be *principal* (or *fixed*) if there exists a nonempty set $A \subseteq S$ such that

$$\mathcal{F} = \{ B \subseteq S \mid A \subseteq B \},$$

in which case we also say $\mathcal{F}$ is the principal filter generated by A. If the singleton filter consisting of all subsets containing a single point $x \in S$ is written as

$$\dot{x} = \{ B \subseteq S \mid x \in B \},$$

then every principal filter is an intersection of such point filters.

A filter $\mathcal{F}$ is called a *free fiter* (or *non-principal filter*) if the intersection of all its members is empty, i.e., $\bigcap_{A \in \mathcal{F}} A = \emptyset$.

Definition 1.2.6 (Ultrafilter on a set)**.** (♠ TODO: AI generated, verify)

Let S be a nonempty set. A filter $\mathcal{U}$ on S is called an *ultrafilter on S* if it is maximal among all filters on S with respect to inclusion. Equivalently, $\mathcal{U}$ is an ultrafilter if and only if for every subset $A \subseteq S$, either $A \in \mathcal{U}$ or $S \setminus A \in \mathcal{U}$.

An ultrafilter $\mathcal{U}$ is called a *principal ultrafilter* (or *fixed ultrafilter*) if there exists a point $x \in S$ such that

$$\mathcal{U} = \{ A \subseteq S \mid x \in A \}.$$

Principal ultrafilters are exactly those of the form $\dot{x}$ where $\dot{x}$ is the point filter at x. An ultrafilter that is not principal is called a *free ultrafilter* (or *non-principal ultrafilter*).

Every filter on a set can be extended to an ultrafilter on that set. This is equivalent to the Boolean prime ideal theorem, which follows from the Axiom of Choice.

Definition 1.2.7 (Convergence of a filter)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $\mathcal{F}$ be a filter on the underlying set of X. We say that the filter $\mathcal{F}$ *converges* to a point $x \in X$ (written $\mathcal{F} \to x$) if every open neighborhood of x belongs to $\mathcal{F}$, i.e.,

$$\mathcal{N}(x) \subseteq \mathcal{F},$$

where $\mathcal{N}(x)$ denotes the collection of all open neighborhoods of x.

The point x is called a *limit* of the filter $\mathcal{F}$. As with nets, in a Hausdorff space every convergent filter has at most one limit.

Filters and nets provide equivalent descriptions of convergence:

1. A net $(x_\alpha)_{\alpha \in D}$ converges to x if and only if the filter generated by its tails $\{\, x_\gamma \mid \gamma \geq \alpha \,\}_{\alpha \in D}$ converges to x.
2. A filter $\mathcal{F}$ converges to x if and only if it is finer than the neighborhood filter $\mathcal{N}(x)$.

Definition 1.2.8 (Cluster point of a filter)**.** (♠ TODO: AI generated, verify)

Let X be a topological space and let $\mathcal{F}$ be a filter on the underlying set of X. A point $x \in X$ is called a *cluster point* (or *accumulation point*) of the filter $\mathcal{F}$ if for every $A \in \mathcal{F}$ and every open neighborhood U of x, the intersection $A \cap U$ is nonempty. Equivalently,

$$\bigcap_{A \in \mathcal{F}} \overline{A} \neq \emptyset,$$

where $\overline{A}$ denotes the closure of A.

A filter $\mathcal{F}$ has a cluster point if and only if there exists an ultrafilter $\mathcal{U}$ finer than $\mathcal{F}$ that converges to some point. In particular, every free ultrafilter on a compact space converges.

Definition 1.2.9 (Filter basis and filter subbasis)**.** (♠ TODO: AI generated, verify)

Let S be a nonempty set. A nonempty collection $\mathcal{B}$ of subsets of S is called a *filter basis on S* if for any $A, B \in \mathcal{B}$, there exists $C \in \mathcal{B}$ such that $C \subseteq A \cap B$. The filter generated by $\mathcal{B}$ is the collection

$$\mathcal{F} = \{\, F \subseteq S \mid \exists B \in \mathcal{B},\ B \subseteq F \,\}.$$

Every filter basis generates a unique filter, and every filter has many different bases.

A nonempty collection $\mathcal{S}$ of subsets of S is called a *filter subbasis on S* if the collection of all finite intersections of members of $\mathcal{S}$ forms a filter basis on S. The filter generated by $\mathcal{S}$ is defined to be the filter generated by this filter basis.

Definition 1.2.10. A topological space X is *disconnected* if there exist two disjoint, non-empty open sets (Definition 1.1.1) U and V such that $X = U \cup V$. The space X is *connected* if it is not disconnected.

Definition 1.2.11. Let X be a topological space and x a point in X. The *connected component of x* is the union of all connected (Definition 1.2.10) subspaces (Definition 1.1.26) of X that contain x.

Definition 1.2.12. Let X be a topological space. X is *totally disconnected* if the connected component (Definition 1.2.11) of every point $x \in X$ consists only of the singleton set $\{x\}$.

Proposition 1.2.13. Let X be a compact (Definition 1.2.14) Hausdorff space (Definition 1.1.25). X is totally disconnected (Definition 1.2.12) if and only if the set of clopen subsets (Definition 1.1.2) of X forms a basis for the topology (Definition 1.1.17) of X.

1.2.2. *Compact topological spaces.*

Definition 1.2.14 (Compact topological space)**.** A topological space $(X, \mathcal{T})$ is *compact* if every open cover of X admits a finite subcover; that is, for every collection $\{U_i\}_{i \in I}$ of open sets in $\mathcal{T}$ such that $X = \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ such that $X = \bigcup_{j=1}^n U_{i_j}$.

Some mathematicians, e.g. algebraic geometers, would refer to this property as *quasi-compactness*.

Here is an alternate, essentially equivalent definition, of compactness:

Definition 1.2.15 (Compactness)**.** Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). A subset $K \subseteq X$ is *compact* if for every collection $\{U_i\}_{i \in I}$ of open sets such that $K \subseteq \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ with $K \subseteq \bigcup_{j=1}^n U_{i_j}$.

Proposition 1.2.16. Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). A subspace (Definition 1.1.26) $K \subseteq X$ is compact in the sense of Definition 1.2.15 if and only if the space K is compact in the sense of Definition 1.2.14.

Definition 1.2.17 (Locally compact)**.** Let $(X, \mathcal{T})$ be a topological space (Definition 1.1.1). X is *locally compact* if for every $x \in X$, there exists an open set $U \in \mathcal{T}$ containing x and a compact set (Definition 1.2.14) $K \subseteq X$ such that $x \in U \subseteq K$.

Definition 1.2.18. Let $(X, \mathcal{T})$ be a topological space and let $A \subseteq X$.

- The *interior of A*, denoted by $\mathrm{int}(A)$, is the union of all open sets contained in A:

$$\mathrm{int}(A) = \bigcup \{U \in \mathcal{T} : U \subseteq A\}.$$

- The *boundary of A*, denoted by ∂A, is defined as

$$\partial A = \overline{A} \setminus \mathrm{int}(A).$$

  (Definition 1.1.31) Equivalently, a point $x \in X$ belongs to ∂A if every open neighborhood of x intersects both A and its complement $X \setminus A$.

1.3. **Continuous maps of topological spaces.**

Definition 1.3.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition 1.1.1). A map $f : X \to Y$ is called *continuous* if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X, that is,

$$\forall V \in \mathcal{T}_Y,\ f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X.

A *map of topological spaces* usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by $C(X, Y)$. Other standard notation include $\mathrm{Hom}_{\mathrm{Top}}(X, Y)$ or $\mathrm{Top}(X, Y)$ coming from more general notation for morphisms between objects in a category (Definition B.0.1).

Definition 1.3.2. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition 1.1.1). A function $f : X \to Y$ is called a *homeomorphism* if it satisfies all of the following:

1. f is bijective (Definition A.0.2);
2. f is continuous (Definition 1.3.1) with respect to $\mathcal{T}_X$ and $\mathcal{T}_Y$; and
3. the inverse map $f^{-1} : Y \to X$ (Definition A.0.2) is also continuous.

(♠ TODO: equivalence class) If such a function exists, the spaces X and Y are said to be *homeomorphic*, or that X and Y have the *same topological type* — accordingly, the *topological type of a topological spaces* refers to its equivalence under the equivalence relation of homeomorhpic spaces.

Definition 1.3.3 (Pointed topological space)**.** Let X be a topological space (Definition 1.1.1) and let $x_0 \in X$ be a chosen element of X. A *pointed/based (topological) space* is a pair (X, x_0) consisting of the space X together with the distinguished point x_0, called the *base point of X*. If the base point of a pointed space (X, x_0) is understood, then it may be suppressed from notation; in particular, X may be written as a pointed space as opposed to the full notation of (X, x_0).

A *morphism of pointed spaces* (or *based map*) or *continuous map* between pointed spaces (X, x_0) and (Y, y_0) is a continuous map (Definition 1.3.1)

$$f : X \to Y$$

such that $f(x_0) = y_0$.

The collection of pointed spaces with their morphisms form a locally small (Definition B.0.2) category (Definition B.0.1), often called the *category of pointed spaces*. This category is often denoted by notations such as Top_*, $\mathrm{Top}_\bullet$, $\mathbf{Top}_*$, $\mathbf{Top}_\bullet$, etc. The set of continuous maps from pointed spaces X to Y may denoted by notations such as $C_*(X, Y)$, $C_\bullet(X, Y)$, $\mathrm{Top}_*(X, Y)$, $\mathrm{Top}_\bullet(X, Y)$, $\mathrm{Hom}_{\mathrm{Top}_\bullet}(X, Y)$, etc.

Definition 1.3.4 (Preimage of a subset under a map of sets)**.** Let X and Y be sets and let $f : X \to Y$ be a function. Let $B \subseteq Y$ be a subset of the codomain Y. The *preimage of B* *under f* (also called the *inverse image of B under f*) is the subset of X defined by

$$f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X.$$

If B is a singleton set $\{y\}$, then we often denote $f^{-1}(B)$ by $f^{-1}(y)$.

Definition 1.3.5 (Fiber of a map of topological spaces)**.** Let X and Y be topological spaces (Definition 1.1.1) and let $f : X \to Y$ be a continuous map (Definition 1.3.1). For a point $y \in Y$, the *fiber of f over y* is the inverse image (Definition 1.3.4) $f^{-1}(y) = f^{-1}(\{y\})$ endowed with the subspace topology (Definition 1.1.26) induced from X. The fiber is also denoted by notations such as $\mathrm{Fib}_f(y)$ or X_y.

1.4. **Constructions of topological spaces from other topological spaces.**

Definition 1.4.1 (Product of Sets)**.** Let I be a (possibly infinite (Definition A.0.1) but small) index set and let $\{A_i\}_{i\in I}$ be a family of sets indexed by I. The *Cartesian product of the family $\{A_i\}_{i\in I}$*, denoted by $\prod_{i\in I} A_i$, is defined as the set of all tuples/functions

$$\prod_{i\in I} A_i := \{(a_i)_{i\in I} \mid a_i \in A_i \text{ for all } i \in I\},$$

where $(a_i)_{i\in I}$ denotes a function from I to $\bigcup_{i\in I} A_i$ such that $(a_i)_{i\in I}(i) = a_i \in A_i$ for each $i \in I$.

The self product of a set A indexed by I is often denoted by A^I. Note that elements of A^I can be identified with functions $I \to A$. The finite self product of A taken n times is often denoted by A^n. For finitely many sets $A_1, \ldots, A_n$, their Cartesian product is denoted by $A_1 \times \cdots \times A_n$. Elements of such a finite product may be written as $(a_1, \ldots, a_n)$.

Definition 1.4.2. Let I be an index set and $\{X_i\}_{i\in I}$ be a collection of topological spaces. Let $X = \prod_{i\in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

The *box topology on X* is the topology generated by (Definition 1.1.17) the base (Definition 1.1.17) consisting of all sets of the form $\prod_{i\in I} U_i$, where each U_i is an open subset (Definition 1.1.1) of X_i.

Definition 1.4.3. Let I be an index set and $\{X_i\}_{i\in I}$ be a collection of topological spaces. Let $X = \prod_{i\in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

The *product topology* (or *Tychonoff topology)* on X is the topology generated by (Definition 1.1.17) the base (Definition 1.1.17) consisting of all sets of the form $\prod_{i\in I} U_i$, where each U_i is an open subset (Definition 1.1.1) of X_i and $U_i = X_i$ for all but finitely many indices $i \in I$.

The *product* or the *product space of the X_i* refers to X equipped with the product topology.

For each $j \in I$, let $p_j : \prod_{i\in I} X_i \to X_j$ denote the canonical projection map. The product topology is the coarsest topology (Definition 1.1.3) on X such that all projection maps p_j are continuous (Definition 1.3.1).

Proposition 1.4.4. Let I be an index set and $\{X_i\}_{i\in I}$ be a collection of topological spaces. Let $X = \prod_{i\in I} X_i$ be the set theoretic Cartesian product (Definition 1.4.1) of these spaces.

If $X = \prod_{i \in I} X_i$ is endowed with the product topology (Definition 1.4.3), then a function $f : Z \to X$ from a topological space (Definition 1.1.1) Z is continuous (Definition 1.3.1) if and only if each component function $f_i = p_i \circ f : Z \to X_i$ is continuous.

This universal property generally fails for the box topology (Definition 1.4.2) when I is infinite (Definition A.0.1).

Theorem 1.4.5 (Tychonoff's theorem)**.** Let I be an index set and let $\{X_i\}_{i \in I}$ be a collection of compact spaces (Definition 1.2.14).

The product space (Definition 1.4.2) $X = \prod_{i \in I} X_i$ is compact in the product topology (Definition 1.1.1).

1.5. **Concrete examples of topological spaces.**

Definition 1.5.1. For a positive integer n, let $\mathbb{R}^n$ denote the n-fold Cartesian product of the real line $\mathbb{R}$ with itself:

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R} \text{ for all } i = 1, \ldots, n\}.$$

The set $\mathbb{R}^n$ is called *Euclidean n-space*. A point $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ is associated with the *Euclidean norm*

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

The corresponding metric (Definition 1.7.1) $d : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$ is given by

$$d(x, y) = \|x - y\|.$$

This metric induces (Definition 1.7.4) the standard topology on $\mathbb{R}^n$, called the *Euclidean topology*.

The Euclidean topology on $\mathbb{R}^n$ coincides with the product topology (Definition 1.4.3) of $\prod_{i=1}^n \mathbb{R}$ where each copy of $\mathbb{R}$ is eqiupped with its Euclidean topology.

Definition 1.5.2 (Interval in the real line)**.** Let $(\mathbb{R}, \leq)$ be the field of real numbers equipped with its usual total order. A subset $I \subseteq \mathbb{R}$ is called an *interval in the real line* if for all $x, y, z \in \mathbb{R}$ one has

$$x < z < y \text{ and } x \in I,\ y \in I \implies z \in I.$$

Equivalently, I is an interval if whenever $x, y \in I$ with $x < y$, then every $z \in \mathbb{R}$ satisfying $x < z < y$ also lies in I. By convention, the empty set $\varnothing$ and singletons $\{x\}$ for $x \in \mathbb{R}$ are also considered intervals, since they satisfy this property.

Let $a, b \in \mathbb{R}$ with $a \leq b$. We use the following standard notations for intervals in $\mathbb{R}$:

$$\begin{aligned}
(a,b) &= \{x \in \mathbb{R} : a < x < b\}, \\
[a,b] &= \{x \in \mathbb{R} : a \leq x \leq b\}, \\
(a,b] &= \{x \in \mathbb{R} : a < x \leq b\}, \\
[a,b) &= \{x \in \mathbb{R} : a \leq x < b\}, \\
(a,\infty) &= \{x \in \mathbb{R} : a < x\}, \\
[a,\infty) &= \{x \in \mathbb{R} : a \leq x\}, \\
(-\infty,b) &= \{x \in \mathbb{R} : x < b\}, \\
(-\infty,b] &= \{x \in \mathbb{R} : x \leq b\}.
\end{aligned}$$

We also regard $\mathbb{R}$ itself and the empty set $\varnothing$ as intervals when convenient.

An interval of the form (a, b) for $a \in \mathbb{R} \cup \{infty\}$ and $b \in \mathbb{R} \cup \{\infty\}$ is called an *open interval*. An interval of the form $[a, b]$ for $a \in \mathbb{R}$ and $b \in \mathbb{R}$ is called a *closed interval*. An interval of the form $(a, b]$ for $a \in \mathbb{R} \cup \{infty\}$ and $b \in \mathbb{R}$ or of the form $[a, b)$ for $a \in \mathbb{R}$ and $b \in \mathbb{R} \cup \{\infty\}$ is called a *half open interval*.

Definition 1.5.3. (♠ TODO: divide into open and closed) Let $n \geq 0$ be an integer. The *n-dimensional (unit) disk (or (unit) ball)*, often denoted by D^n (or B^n), is the subspace (Definition 1.1.26) of $\mathbb{R}^n$ (Definition 1.9.1) defined by:

$$D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$$

For $n = 1$, the disk D^1 is the closed interval (Definition 1.5.2) $I = [-1, 1]$, which is homeomorphically (Definition 1.3.2) to the standard unit interval $[0, 1]$.

One also often means by an *n-dimensional disk* any space homeomorphic to D^n.

Definition 1.5.4. Let $n \geq 0$ be an integer. The *n-dimensional (unit) sphere*, denoted by S^n, is the subspace of $\mathbb{R}^{n+1}$ defined by:

$$S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$$

The sphere S^n is the boundary (Definition 1.2.18) of the $(n+1)$-dimensional disk D^{n+1} (Definition 1.5.3). Specifically, S^0 consists of the two discrete points $\{-1, 1\}$.

S^1 is called the *unit circle*.

One also often means by an *n-dimensional sphere* any space homeomorphic to S^n.

Definition 1.5.5. The *standard unit interval* or the *closed unit interval*, denoted by I, is the subspace (Definition 1.1.26) of $\mathbb{R}$ (Definition 1.9.1) defined by:

$$I = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

(Definition 1.5.2)

Definition 1.5.6. The *n-dimensional unit cube* is the n-fold Cartesian product (Definition 1.4.1) $I^n = I \times \cdots \times I$, equipped with the product topology (Definition 1.4.3) where I is the standard unit interval (Definition 1.5.5) equipped with its usual topology. This topology on I^n coincides with the subspace topology (Definition 1.1.26) of I^n as a subset of $\mathbb{R}^n$ (Definition 1.9.1).

Definition 1.5.7. Let $n \geq 1$ be an integer. The *n-dimensional torus*, denoted by T^n, is the topological space (Definition 1.1.1) defined as the n-fold Cartesian product (Definition 1.4.1) of the circle (Definition 1.5.4):

$$T^n = \underbrace{S^1 \times S^1 \times \cdots \times S^1}_{n \text{ times}}$$

(♠ TODO: quotient space by a group action) Equivalently, T^n is the quotient space $\mathbb{R}^n/\mathbb{Z}^n$ under the action of translation by the integer lattice.

Definition 1.5.8. Let $n \geq 0$ be an integer and let $\mathbb{K}$ be the field of real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$. The *n-dimensional projective space over* $\mathbb{K}$, denoted by $\mathbb{K}P^n$, is the set of 1-dimensional vector subspaces of $\mathbb{K}^{n+1}$, equipped with the quotient topology induced by the map:

$$q : \mathbb{K}^{n+1} \setminus \{0\} \to \mathbb{K}P^n$$

which sends a non-zero vector to the line it spans. More explicitly, elements of $\mathbb{K}P^n$ are denoted by $[z_0 : z_1 : \cdots : z_n]$ where $z_0, \ldots, z_n$ are elements of $\mathbb{K}$, not all zero, and where $[z_0 : z_1 : \cdots : z_n] = [\lambda z_0 : \lambda z_1 : \cdots : \lambda z_n]$ for any $\lambda \in \mathbb{K} \setminus \{0\}$.

Explicitly, for $\mathbb{K} = \mathbb{R}$, this is the real projective space $\mathbb{R}P^n$; for $\mathbb{K} = \mathbb{C}$, this is the complex projective space $\mathbb{C}P^n$.

1.6. Compactly generated.

Definition 1.6.1. (♠ TODO: final topology) A topological space (Definition 1.1.1) X is said to be *compactly generated* (or a *k-space*) if a subset $U \subseteq X$ is open whenever for every compact (Definition 1.2.14) subset $K \subseteq X$, the intersection $U \cap K$ is open in the subspace K. Equivalently, X is compactly generated if and only if the topology of X is the final topology with respect to the collection of inclusions $K \hookrightarrow X$ for compact $K \subseteq X$.

Definition 1.6.2. A topological space (Definition 1.1.1) X is said to be *weakly Hausdorff* if for every continuous map (Definition 1.3.1) $f : K \to X$ from a compact (Definition 1.2.14) Hausdorff (Definition 1.2.15) space K, the image $\mathrm{Im}(f)$ is closed in X. Equivalently, X is weakly Hausdorff if and only if for every compact Hausdorff K, the induced map $f : K \to X$ is a closed map.

1.7. Metrics, absolute values, and norms.

1.7.1. *Extended metric spaces.*

Definition 1.7.1 (Extended metric on a set valued in an ordered semiring with adjoined infinity)**.** Let M be a set and let T be an ordered commutative monoid. Let T^∞ be the ordered commutative monoid with adjoined infinity obtained from T.

An *extended metric on M valued in T^∞* is a function

$$d : M \times M \to T^\infty$$

such that for all $x, y, z \in M$:

1. **Non-negativity and identity of indiscernibles:** $d(x, y) \geq 0_T$, and $d(x, y) = 0_T$ if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$,

where addition and order are those of T^∞, and the convention that sums involving ∞ behave so that $t + \infty = \infty$ for any $t \in T^\infty$. A set equipped with an extended metric valued in T^∞ is called an *extended metric space valued in T^∞*.

If the range of d is contained in T, then we call it a *metric on M valued in T*, with M called a *metric space valued in T*.

An *extended metric on M* without reference to a valued a specific choice of T almost usually refers to the standard notion of extended metric on M valued in $T = \mathbb{R}_{\geq 0} = [0, \infty]$.

An extended metric space has a natural topological space (Definition 1.1.1) structure (Definition 1.7.4).

Definition 1.7.2 (Isometry for extended metric spaces)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X, d_X) and (Y, d_Y) be extended metric spaces valued in T^∞ (Definition 1.7.1). A map $f : X \to Y$ is called an *isometry* if for all $x_1, x_2 \in X$,

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

An isometry is automatically injective. If, in addition, f is surjective, it is called an *isometric isomorphism*, and X and Y are said to be *isometric*.

Definition 1.7.3 (Extended metric induced by an extended norm on a module over a commutative ring with absolute value)**.** Let T be an ordered semiring, let R be a commutative ring (Definition D.0.1) equipped with an absolute value valued in T (Definition 1.8.5):

$$|\cdot| : R \to T,$$

and let M be a module over R. Let T^∞ be the ordered semiring with adjoined infinity obtained from T, and let

$$\|\cdot\| : M \to T^\infty$$

be an extended norm on M valued in T^∞.

The *extended metric induced by the extended norm* is the function

$$\boxed{d : M \times M \to T^\infty, \quad (x, y) \mapsto \|x - y\|}$$

defined by

$$d(x, y) := \|x - y\|.$$

The map d satisfies the axioms of an extended metric valued in T^∞: for all $x, y, z \in M$,

1. Positive definiteness: $d(x, y) = 0_T$ if and only if $\|x - y\| = 0_T$ if and only if $x - y = 0_M$ if and only if $x = y$.
2. Symmetry: $d(x, y) = \|x - y\| = \|(-1_R)(y - x)\| = |-1_R| \cdot \|y - x\| = \|y - x\| = d(y, x)$, where $|-1_R| = 1_T$ follows from multiplicativity of the absolute value on the ring R.

3. Triangle inequality: $d(x,z) = \|x-z\| = \|(x-y)+(y-z)\| \leq \|x-y\| + \|y-z\| = d(x,y)+d(y,z)$, by the triangle inequality for $\|\cdot\|$ with addition in T^∞.

If the range of $\|\cdot\|$ is contained in T (so that $\|\cdot\|$ is a norm rather than an extended norm), then d takes values in T and is a metric.

In the case that $R = F$ is a field and $M = V$ is a vector space over F, we recover the standard definition of an extended metric on V induced by a norm valued in T^∞. In particular, when $T = \mathbb{R}_{\geq 0}$, we recover the standard definition of a metric/extended metric on V taking values in $\mathbb{R}_{\geq 0}$ or $[0,\infty]$.

Definition 1.7.4 (Topology induced by an extended metric valued in an ordered monoid with adjoined infinity)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X,d) be an extended metric space valued in T^∞ (Definition 1.7.1).

1. Given $\varepsilon \in T_{>0}$ and $x \in X$, the set
$$B(x,\varepsilon) := \{y \in X : d(x,y) < \varepsilon\}$$
is called the *open ball of radius ε centered at x*.
2. The *topology induced by d on X* is the topology (Definition 1.1.1) defined by declaring a subset $U \subseteq X$ to be open if for every $x \in U$, there exists $\varepsilon \in T_{>0}$ such that $B(x,\varepsilon) \subseteq U$. In particular, $B(x,\varepsilon)$ is an open set for this topology for all $x \in X$ and $\varepsilon \in T_{>0}$.

1.8. **Convergence in extended metric spaces.**

Definition 1.8.1 (Convergence and limits in an extended metric space valued in an ordered monoid with adjoined infinity)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X,d) be an extended metric space valued in T^∞ (Definition 1.7.1). A sequence $(x_n)_{n=1}^\infty$ in X *converges to a point $x \in X$* if for every $\varepsilon \in T_{>0}$, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$,
$$d(x_n, x) < \varepsilon.$$
In that case, x is called the *limit of the sequence (x_n)*, and we write
$$\lim_{n\to\infty} x_n = x \quad \text{or} \quad x_n \to x$$

When $T = \mathbb{R}_{\geq 0}$, this recovers the standard definition of convergence in an extended metric space. In general, if T has no minimum positive element (e.g., $T = \mathbb{R}_{\geq 0}$ or any dense ordered semiring), then the condition "$d(x_n,x) < \varepsilon$ for all $\varepsilon \in T_{>0}$" is strictly weaker than $d(x_n,x) = 0_T$, allowing sequences to approach x without eventually equaling it. If T has a minimum positive element, then convergence forces eventual equality: (x_n) converges to x if and only if $x_n = x$ for all sufficiently large n.

Definition 1.8.2 (Cauchy sequence in an extended metric space valued in an ordered monoid with adjoined infinity)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X,d) be an extended metric space valued in T^∞ (Definition 1.7.1). A sequence $(x_n)_{n\in\mathbb{N}}$ in X is called a *Cauchy sequence* if for every $\varepsilon \in T_{>0}$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$,

$$d(x_m, x_n) < \varepsilon.$$

Because the quantification is over $\varepsilon \in T_{>0} \subseteq T$ (not including ∞), any Cauchy sequence must eventually have finite pairwise distances: for all sufficiently large m, n, we have $d(x_m, x_n) \in T$.

When $T = \mathbb{R}_{\geq 0}$, this recovers the standard definition of a Cauchy sequence. In general, if T is dense at zero, then the condition allows pairwise distances to become arbitrarily small without being zero. If T has a minimum positive element ε_0, then every Cauchy sequence is eventually constant: for all sufficiently large m, n, we must have $d(x_m, x_n) < \varepsilon_0$. Since ε_0 is the smallest strictly positive element, this forces $d(x_m, x_n) = 0_T$ and hence $x_m = x_n$.

Definition 1.8.3 (Complete extended metric space valued in an ordered monoid with adjoined infinity)**.** Let T be an ordered commutative monoid, let T^∞ be the ordered monoid with adjoined infinity obtained from T, and let (X, d) be an extended metric space valued in T^∞ (Definition 1.7.1). The space (X, d) is said to be *complete* if every Cauchy sequence in X (Definition 1.8.2) converges to a point in X (Definition 1.8.1); that is, for every sequence $(x_n)_{n\in\mathbb{N}}$ in X such that for all $\varepsilon \in T_{>0}$ there exists $N \in \mathbb{N}$ with $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$, there exists $x \in X$ such that

$$\lim_{n\to\infty} d(x_n, x) = 0_T.$$

When $T = \mathbb{R}_{\geq 0}$, this recovers the standard definition of a complete extended metric space. Note that if T has a minimum positive element, then every Cauchy sequence is eventually constant (see Definition 1.8.2), so completeness holds trivially for all such spaces.

Proposition 1.8.4. Let T be an ordered semiring, let T^∞ be the ordered semiring with adjoined infinity obtained from T, and let (X, d) be an extended metric space valued in T^∞ (Definition 1.7.1). A completion $(\widehat{X}, \widehat{d})$ of (X, d) exists and is unique up to unique isometry. Moreover, if (X, d) is a metric space, then so is $(\widehat{X}, \widehat{d})$.

Proof. (♠ TODO: :) actually complete this proof Let $\mathcal{C}$ be the set of all Cauchy sequences in (X, d):

$$\mathcal{C} = \{(x_n)_{n\in\mathbb{N}} : (x_n) \text{ is a Cauchy sequence in } (X, d)\},$$

and define a relation $\sim$ on $\mathcal{C}$ by

$$(x_n) \sim (y_n) \iff \lim_{n\to\infty} d(x_n, y_n) = 0.$$

This relation is well-defined and in fact an equivalence relation. (♠ TODO: Show that the relation is well-defined and an equivalence relation) Let $\widehat{X} = \mathcal{C}/\sim$ and let $\widehat{d} : \widehat{X} \times \widehat{X} \to [0, \infty]$ be defined by

$$\widehat{d}([(x_n)], [(y_n)]) := \lim_{n\to\infty} d(x_n, y_n).$$

This limit exists as a value in $[0, \infty]$ and is independent of representative sequences chosen. (♠ TODO: Demonstrate why the limit exists and is well defined)

It can be verified that $\widehat{d}$ is an extended metric on $\widehat{X}$ and that $(\widehat{X}, \widehat{d})$ is complete by construction.

Define an embedding

$$i : X \to \widehat{X}, \quad i(x) := [(x, x, x, \ldots)],$$

which is an isometry:

$$\widehat{d}(i(x), i(y)) = \lim_{n\to\infty} d(x, y) = d(x, y).$$

Moreover, $i(X)$ is dense in $\widehat{X}$ because any equivalence class $[(x_n)]$ of a Cauchy sequence can be approximated arbitrarily closely by points $i(x_n)$ in the image of X.

Uniqueness:

Suppose $(\widehat{X}', \widehat{d}')$ is another completion of (X, d) with isometric embedding $i' : X \to \widehat{X}'$.

By the universal property of completions, there exists a unique isometry

$$\Phi : \widehat{X} \to \widehat{X}'$$

such that $\Phi \circ i = i'$.

The map Φ is onto since $i'(X)$ is dense in $\widehat{X}'$ and Φ preserves limits of Cauchy sequences.

Therefore, Φ is a unique isometry between the completions, and the completion is unique up to unique isometry. $\square$

1.8.1. *Absolute values on fields.*

Definition 1.8.5 (Absolute value on a ring)**.** Let R be a commutative ring (Definition D.0.1) and let T be an ordered semiring.

An *absolute value on R valued in T* (or a *norm on R valued in T*) is a function

$$|\cdot| : R \to T$$

that satisfies the following properties for all $a, b \in R$:

1. Positive definiteness: $|a| = 0_T$ if and only if $a = 0_R$.
2. Multiplicativity: $|ab| = |a| \cdot |b|$,
3. Subadditivity/triangle inequality: $|a + b| \leq |a| + |b|$,

A ring equipped with an absolute value may be called a *normed ring* or an *absolute valued ring*. A field equipped with an absolute value may be called a *normed field*. It may also be called a *valued field*, but this should not be confused with the more algebraic notion of a field equipped with a valuation with values in a more general ordered group.

In the case that T is not explicit, the absolute value $|\cdot|$ is almost always taken to be valued in $\mathbb{R}_{\geq 0}$.

1.8.2. *Norms on vector spaces over fields with absolute values.*

Definition 1.8.6 (Extended norm on a vector space over a field with absolute value)**.** Let T be an ordered semiring, let F be a field equipped with an absolute value valued in T (Definition 1.8.5):

$$|\cdot| : F \to T,$$

and let V be a vector space over F. Let T^∞ be the ordered semiring with adjoined infinity obtained from T.

An *extended norm on V valued in T^∞* is a function

$$\|\cdot\| : V \to T^\infty$$

satisfying for all $x, y \in V$ and all scalars $\alpha \in F$:

1. Positive definiteness: $\|x\| = 0_T$ if and only if $x = 0$.
2. Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$, where the multiplication on the right is in T^∞.
3. Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$, where addition and order are those of T^∞.

A vector space with an extended norm valued in T^∞ over a field with absolute value is called an *extended normed (vector) space*.

If the range of the extended norm is contained in T, then the extended norm is a *norm* in the usual sense and V may be called a *normed vector space* or a *normed space*.

In the case that $T = \mathbb{R}_{\geq 0}$, we recover the standard definition of an extended norm taking values in $\mathbb{R}_{\geq 0}$ or $[0, \infty]$.

Definition 1.8.7 (Topology induced by a norm on a module over a commutative ring with absolute value)**.** Let T be an ordered semiring, let R be a commutative ring (Definition D.0.1) equipped with an absolute value valued in T (Definition 1.8.5):

$$|\cdot| : R \to T,$$

and let M be a module over R. Let T^∞ be the ordered semiring with adjoined infinity obtained from T, and let $\|\cdot\|$ be an extended norm on M valued in T^∞. The *topology induced by the extended norm $\|\cdot\|$ on M* is defined by declaring a subset $U \subseteq M$ to be open if for every $x \in U$, there exists $\varepsilon \in T_{>0}$ such that

$$B(x, \varepsilon) := \{y \in M : \|y - x\| < \varepsilon\}$$

is contained in U. The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x*.

In the case that $R = F$ is a field and $M = V$ is a vector space over F, we recover the standard definition of the topology induced by a norm on a vector space.

Equivalently, the topology on M induced by the extended norm $\|\cdot\|$ is the topology on M induced by (Definition 1.7.4) the extended metric $d : M \times M \to T^\infty$ valued in T^∞ induced by $\|\cdot\|$ (Definition 1.7.3).

1.8.3. *Inner product on a vector space over the reals or complex numbers.*

Definition 1.8.8. An *inner product on a vector space V over a field $\mathbb{F}$* (either $\mathbb{R}$ or $\mathbb{C}$) is a function

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

satisfying, for all $u, v, w \in V$ and all scalars $\alpha \in \mathbb{F}$:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$
2. **Linearity in the first argument:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$
3. **Positive-definiteness:** $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ if and only if $v = 0$

A vector space over $\mathbb{F}$ equipped with such an inner product is called an *inner product space*.

Definition 1.8.9. Let V be a vector space over a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $\langle \cdot, \cdot \rangle$ be an inner product (Definition 1.8.8). The *norm induced by $\langle \cdot, \cdot \rangle$* is the norm (Definition 1.8.6)

$$\| \cdot \| : V \to \mathbb{R}_{\geq 0}$$

defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

for any $v \in V$.

1.9. **Euclidean space.**

Definition 1.9.1. For a positive integer n, let $\mathbb{R}^n$ denote the n-fold Cartesian product of the real line $\mathbb{R}$ with itself:

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R} \text{ for all } i = 1, \ldots, n\}.$$

The set $\mathbb{R}^n$ is called *Euclidean n-space*. A point $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ is associated with the *Euclidean norm*

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

The corresponding metric (Definition 1.7.1) $d : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$ is given by

$$d(x, y) = \|x - y\|.$$

This metric induces (Definition 1.7.4) the standard topology on $\mathbb{R}^n$, called the *Euclidean topology*.

The Euclidean topology on $\mathbb{R}^n$ coincides with the product topology (Definition 1.4.3) of $\prod_{i=1}^n \mathbb{R}$ where each copy of $\mathbb{R}$ is eqiupped with its Euclidean topology.

1.10. **Connectivity of topological spaces.**

Definition 1.10.1. Let X be a topological space. It is said to be *path connected* if for any two points x_0, x_1 in X, there is some path (Definition 3.0.7) $\gamma : [0, 1] \to X$ in X such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

Definition 1.10.2. Let X be a topological space. It is said to be *locally path connected* if for any $x \in X$, there is an open neighborhood (Definition 1.1.16) $x \in U \subseteq X$ that is path connected (Definition 1.10.1).

Definition 1.10.3. Let X be a topological space.

Let $x \in X$. The *path component of x in X* is the maximal path connected subspace of X containing x.

The set of path components in X is denoted $\pi_0(X)$. For any $x \in X$, $\pi_0(X)$ coincides, as a set, with the zeroth homotopy set (Definition 3.0.8) $\pi_0(X, x_0)$ of the pointed space (Definition 1.3.3) (X, x_0).

Definition 1.10.4. A topological space X is *simply connected* if it is path-connected (Definition 1.10.1) and its fundamental group (Definition 3.0.8) $\pi_1(X, x_0)$ is trivial for some (and hence every) base (Definition 1.1.17) point $x_0 \in X$. Equivalently, X is simply connected if any two continuous maps (Definition 1.3.1) $f, g : [0, 1] \to X$ with $f(0) = g(0)$ and $f(1) = g(1)$ are homotopic relative to (Definition 3.0.1) $\{0, 1\}$.

Proposition 1.10.5. Let X be a topological space. X is simply connected (Definition 1.10.4) if and only if it is path-connected (Definition 1.10.1) and every continuous map (Definition 1.3.1) $S^1 \to X$ from the unit circle into X is null-homotopic.

Proposition 1.10.6. (♠ TODO:) Let X be a topological space. If X is contractible, then X is simply connected. The converse is not true in general; for example, the n-dimensional sphere S^n is simply connected for $n \geq 2$ but is not contractible.

Definition 1.10.7. A topological space X is *locally simply connected* if it admits a basis (Definition 1.1.17) of open sets (Definition 1.1.1) $\{U_\alpha\}$ such that each U_α is simply connected (Definition 1.2.10). That is, for every $x \in X$ and every open neighborhood (Definition 1.1.16) V of x, there exists a simply connected open neighborhood $U \subset V$ of x.

Proposition 1.10.8. Let X be a topological space. If X is locally simply connected (Definition 1.10.7), then X is both locally path-connected (Definition 1.10.2) and semi-locally simply connected (Definition 1.10.11).

Proposition 1.10.9. Every topological manifold, with or without boundary, is locally simply connected (Definition 1.10.7) because every point has a neighborhood homeomorphic to an open ball in $\mathbb{R}^n$ or $\mathbb{H}^n$, both of which are simply connected.

Definition 1.10.10. (♠ TODO: contractible) Let X be a topological space. X is *locally contractible* if every point $x \in X$ has a basis (Definition 1.1.17) of contractible open neighborhoods (Definition 1.1.16). Since every contractible space is simply connected, every locally contractible space is locally simply connected (Definition 1.10.7). (♠ TODO: simply connected space)

Definition 1.10.11. A topological space X is *semi-locally simply connected* if for every point $x \in X$, there exists an open neighborhood (Definition 1.1.16) U of x such that the inclusion-induced homomorphism (Definition C.0.2) on fundamental groups (Definition 3.0.8)

$$i_* : \pi_1(U, x) \to \pi_1(X, x)$$

is the trivial homomorphism.

Proposition 1.10.12. Let X be a topological space. The condition of semi-local simple connectivity (Definition 1.10.11) is equivalent to the requirement that every point $x \in X$ has an open neighborhood U such that every loop in U based at x is null-homotopic in X. Note that the loops are not required to be null-homotopic within U itself. (♠ TODO: null-homotopic)

1.11. **Stone spaces, ultrafilters, and Boolean algebras.**

Definition 1.11.1. A topological space X is a *Stone space* if it is compact (Definition 1.2.14), Hausdorff (Definition 1.1.25), and totally disconnected (Definition 1.2.12).

Definition 1.11.2. The *category of Stone spaces* is the category (Definition B.0.1) denoted by Stone whose objects are Stone spaces (Definition 1.11.1) (compact (Definition 1.2.14), Hausdorff (Definition 1.1.25), and totally disconnected (Definition 1.2.12) topological spaces (Definition 1.1.1)) and the morphisms are continuous maps (Definition 1.3.1) between them.

Definition 1.11.3. A *Boolean algebra* is a set B equipped with two binary operations $\vee$ (*join*) and $\wedge$ (*meet*), a unary operation $\neg$ (*complement*), and two distinct elements 0 (*bottom*) and 1 (*top*), such that for all $x, y, z \in B$:

- $x \vee (y \vee z) = (x \vee y) \vee z$ and $x \wedge (y \wedge z) = (x \wedge \wedge z)$ (associativity).
- $x \vee y = y \vee x$ and $x \wedge y = y \wedge x$ (commutativity).
- $x \wedge (x \vee y) = x$ and $x \vee (x \wedge y) = x$ (absorption).
- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ (distributivity).
- $x \vee (\neg x) = 1$ and $x \wedge (\neg x) = 0$ (complements).

Definition 1.11.4. Let A and B be Boolean algebras (Definition 1.11.3). A *morphism of Boolean algebras* is a function $f : A \to B$ such that for all $x, y \in A$:

- $f(x \vee y) = f(x) \vee f(y)$,
- $f(x \wedge y) = f(x) \wedge f(y)$,
- $f(\neg x) = \neg f(x)$,
- $f(0_A) = 0_B$ and $f(1_A) = 1_B$.

Definition 1.11.5. The *category of Boolean algebras* is the category (Definition B.0.1) whose objects are Boolean algebras (Definition 1.11.3) and the morphisms are Boolean algebra morphisms (Definition 1.11.4). It is often denoted by Bool.

Proposition 1.11.6. Let A and B be Boolean algebras (Definition 1.11.3). (♠ TODO: corresponding boolean rings) A function $f : A \to B$ is a morphism of Boolean algebras (Definition 1.11.4) if and only if it is a ring homomorphism between the corresponding Boolean rings.

Definition 1.11.7. Let B be a Boolean algebra (Definition 1.11.3). A *filter F on B* is a non-empty subset $F \subseteq B$ such that $0 \notin F$, if $x, y \in F$ then $x \wedge y \in F$, and if $x \in F$ and $x \leq y$ (where $x \leq y$ is defined as $x \wedge y = x$) then $y \in F$. An *ultrafilter* is a filter U that is maximal with respect to set inclusion.

Proposition 1.11.8. Let B be a Boolean algebra and $U \subseteq B$ a filter. U is an ultrafilter if and only if for every $x \in B$, either $x \in U$ or $\neg x \in U$.

Definition 1.11.9. Let B be a Boolean algebra (Definition 1.11.3). The *Stone space of B*, denoted by *$Spec(B)$*, is the set of all ultrafilters (Definition 1.11.7) on B equipped with the topology generated by (Definition 1.1.17) the basis (Definition 1.1.17) of sets

$$U_a = \{p \in Spec(B) : a \in p\}$$

for all $a \in B$.

Theorem 1.11.10 (Stone's representation theorem)**.** Stone's representation theorem states that there is a contravariant equivalence of categories (a duality) between the category of Boolean algebras (Definition 1.11.5) Bool and the category of Stone spaces (Definition 1.11.2) Stone. This equivalence is realized by the following pair of contravariant functors:

- $\mathcal{S} : \mathsf{Bool} \to \mathsf{Stone}$, which maps a Boolean algebra (Definition 1.11.3) B to its Stone space $Spec(B)$ of ultrafilters (Definition 1.11.7).
- $\mathcal{C} : \mathsf{Stone} \to \mathsf{Bool}$, which maps a Stone space X to its Boolean algebra of clopen subsets (Definition 1.1.2) $\mathsf{Clopen}(X)$.

For any Boolean algebra B, there is a natural isomorphism $B \cong \mathcal{C}(\mathcal{S}(B))$, and for any Stone space X, there is a natural homeomorphism (Definition 1.3.2) $X \cong \mathcal{S}(\mathcal{C}(X))$.

2. Covering spaces

Definition 2.0.1.

1. Let X be a topological space.
   A *covering space of X* is a pair $(\tilde{X}, p)$ where $\tilde{X}$ is a topological space and $p : \tilde{X} \to X$ is a continuous (Definition 1.3.1) surjective (Definition A.0.2) map such that for every $x \in X$, there exists an open neighborhood (Definition 1.1.16) U of x (called an *evenly covered neighborhood)* for which the preimage $p^{-1}(U)$ is a disjoint union of open sets (Definition 1.1.1) $V_i \subset \tilde{X}$, each of which is mapped homeomorphically (Definition 1.3.2) onto U by the restriction $p|_{V_i}$.
2. Let (X, x_0) be a pointed topological space (Definition 1.3.3). A *(pointed) covering of (X, x_0)* is a pointed topological space (Definition 1.3.3) $(\tilde{X}, \tilde{x}_0)$ along with a covering $p : \tilde{X} \to X$ of X such that p is a map of pointed space (Definition 1.3.3).

Definition 2.0.2.

1. Let X be a topological space.
   A *morphism of covering spaces of X from $p_1 : \tilde{X}_1 \to X$ to $p_2 : \tilde{X}_2 \to X$* is a continuous map (Definition 1.3.1) $f : \tilde{X}_1 \to \tilde{X}_2$ such that $p_2 \circ f = p_1$.
2. Let (X, x_0) be a pointed topological space (Definition 1.3.3). Let $p_1 : (\tilde{X}_1, \tilde{x}_1) \to (X, x_0)$ to $p_2 : (\tilde{X}_2, \tilde{x}_2) \to (X, x_0)$ be pointed coverings (Definition 2.0.1) of (X, x_0). A *morphism $f : p_1 \to p_2$ of the pointed coverings* is a pointed map $f : (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, x_2)$ such that $p_2 \circ f = p_1$ and $f(\tilde{x}_1) = \tilde{x}_2$.

Definition 2.0.3. Let $p : \tilde{X} \to X$ be a covering space (Definition 2.0.1) of a topological space X. A *deck transformation* (or *covering transformation*, or *automorphism*) of p is a homeomorphism (Definition 1.3.2) $f : \tilde{X} \to \tilde{X}$ such that $p \circ f = p$. The set of all deck transformations forms a group under composition, denoted $\mathrm{Aut}(p)$ or $\mathrm{Deck}(p)$.

Definition 2.0.4. Let X be a topological space. A *universal cover of X* is a pair $(\tilde{X}, p)$ where $p : \tilde{X} \to X$ is a covering space (Definition 2.0.1) such that $\tilde{X}$ is simply connected (Definition 1.10.4).

Theorem 2.0.5. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space (Definition 2.0.1) of a path-connected (Definition 1.10.1) and locally path-connected space (Definition 1.10.2) X. The induced map on fundamental groups (Definition 3.0.8) $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$ is an injective group homomorphism (Definition C.0.2).

Definition 2.0.6. Let $p : \tilde{X} \to X$ be a covering space (Definition 2.0.1) of a topological space X. Let $f : Y \to X$ be a continuous map (Definition 1.3.1) of topological spaces.

A *lift of f (via p)* is a continuous map $\tilde{f} : Y \to \tilde{X}$ satisfying $p \circ \tilde{f} = f$.

Similarly, for a pointed covering $p : (\tilde{X}, x_0) \to (X, x_0)$ and a pointed morphism (Definition 1.3.3) $f : (Y, y_0) \to (x, x_0)$, a *lift of f (via p)* is a pointed map $\tilde{f} : (Y, y_0) \to (\tilde{X}, x_0)$ satisfying $p \circ \tilde{f} = f$.

Proposition 2.0.7. Every simply connected (Definition 1.2.10) topological space space is semi-locally simply connected (Definition 1.10.11). Furthermore, every locally simply connected space is semi-locally simply connected. However, there exist semi-locally simply connected spaces, such as the cone over the Hawaiian earring, that are not locally simply connected. *Remark:* (♠ TODO: define locally simply connected).

Definition 2.0.8. A covering space (Definition 2.0.1) $p : \tilde{X} \to X$ of a topological space X is *normal* (or *Galois)* if for every $\tilde{x}_0 \in X_0$, the image of the induced map $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(\tilde{X}, p(\tilde{x}_0))$ on fundamental groups (Definition 3.0.8) is the normal subgroup of $\pi_1(\tilde{X}, p(\tilde{x}_0))$

Theorem 2.0.9. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space (Definition 2.0.1) and let $f : (Y, y_0) \to (X, x_0)$ be a continuous map (Definition 1.3.1) where Y is path-connected (Definition 1.10.1) and locally path-connected (Definition 1.10.2). A lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x}_0)$ exists if and only if the image of the fundamental group (Definition 3.0.8) of (Y, y_0) is contained in the image of the fundamental group of $(\tilde{X}, \tilde{x}_0)$, i.e.

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

If such a lift exists, it is unique.

See also Theorem 2.0.10.

Theorem 2.0.10. Let $p_X : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ and $p_Y : (\tilde{Y}, \tilde{y}_0) \to (Y, y_0)$ be covering spaces of pointed topological spaces (Definition 1.3.3). Let $f : (X, x_0) \to (Y, y_0)$ be a continuous map (Definition 1.3.1), and assume $\tilde{X}$ is path-connected (Definition 1.10.1) and locally path-connected (Definition 1.10.2).

There exists a unique continuous lift $\tilde{f} : (\tilde{X}, \tilde{x}_0) \to (\tilde{Y}, \tilde{y}_0)$ such that $p_Y \circ \tilde{f} = f \circ p_X$ if and only if

$$f_*(p_{X*}(\pi_1(\tilde{X}, \tilde{x}_0))) \subseteq p_{Y*}(\pi_1(\tilde{Y}, \tilde{y}_0))$$

In the specific case where f is the identity on X, this provides the necessary and sufficient condition for the existence of a morphism of covering spaces (pointed or unpointed).

Any such morphism between connected covering spaces is itself a covering map. See also Theorem 2.0.9.

Theorem 2.0.11. Let X be a connected (Definition 1.2.10), locally path-connected (Definition 1.10.2), and semi-locally simply connected (Definition 1.10.11) space with base (Definition 1.1.17) point x_0. The universal cover (Definition 2.0.4) $\tilde{X}$ can be constructed as the space of homotopy classes (Definition 3.0.3) of paths (Definition 3.0.7) starting at x_0:

$$\tilde{X} = \{[\gamma] \mid \gamma : [0,1] \to X, \gamma(0) = x_0\}$$

with the covering map $p : \tilde{X} \to X$ given by $p([\gamma]) = \gamma(1)$. For any connected covering space (Definition 2.0.1) $q : Y \to X$, there exists a covering map $r : \tilde{X} \to Y$ such that $q \circ r = p$.

Proposition 2.0.12. Let X be a connected (Definition 1.2.10), locally path-connected (Definition 1.10.2), and semi-locally simply connected (Definition 1.10.11) space with base point (Definition 1.3.3) x_0. There are bijective (Definition A.0.2) correspondences:

1. Between the set of isomorphism classes of connected (Definition 1.2.10) pointed covering spaces (Definition 2.0.1) $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$, via $((\tilde{X}, \tilde{x}_0), p) \mapsto p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.
2. Between the set of isomorphism classes of connected covering spaces $p : \tilde{X} \to X$ and the set of conjugacy classes of subgroups of $\pi_1(X, x_0)$, via $p \mapsto [p_*(\pi_1(\tilde{X}, \tilde{x}_0))]$. (♠ TODO: conjugacy class)

Theorem 2.0.13. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering with $\tilde{X}$ (and hence X) connected (Definition 1.2.10). The group of deck transformations (Definition 2.0.3) $\mathrm{Aut}(p)$ is isomorphic to the quotient $N(H)/H$, where $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ and $N(H)$ is its normalizer in $\pi_1(X, x_0)$. If the covering is normal (Definition 2.0.8) (i.e., H is a normal subgroup of $\pi_1(X, x_0)$), then:

$$\mathrm{Aut}(p) \cong \pi_1(X, x_0)/p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

For the universal cover, $\mathrm{Aut}(p) \cong \pi_1(X, x_0)$.

3. Homotopy

The notions of homotopy have the following variants:

1. Absolute, on unpointed spaces
2. Relative, on unpointed spaces
3. Absolute, on pointed spaces
4. Relative, on pointed spaces.

Definition 3.0.1 (Homotopy of maps of topological spaces)**.** Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \to Y$.

1. A *homotopy between two maps $f, g \in C(X, Y)$ relative to K* is a continuous map

$$H : X \times [0,1] \to Y$$

such that for all $x \in X$,

$$H(x,0) = f(x), \quad H(x,1) = g(x),$$

and for all $x \in K$ and $t \in [0,1]$,

$$H(x,t) = f(x) = g(x).$$

If such an H exists, we say f and g are *homotopic relative to K*, and we write $f \simeq g$ rel K or $f \simeq_K g$; this is an equivalence relation.

In the case that $K = \emptyset$, a homotopy relative to K is simply called a *homotopy*. We write we write $f \simeq g$ if a homotopy between f and g exists.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X,Y)$ denote the set of all continuous based maps $f : X \to Y$ satisfying $f(x_0) = y_0$.

   A *homotopy of based maps $f, g \in C_*(X,Y)$ relative to K* is a continuous map

$$H : X \times [0,1] \to Y$$

such that for all $x \in X$,

$$H(x,0) = f(x), \quad H(x,1) = g(x),$$

and for all $k \in K$ and $t \in [0,1]$,

$$H(k,t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0,t) = y_0 \quad \text{for all } t \in [0,1].$$

If such an H exists, we say f and g are *based homotopic relative to K*, and we write $f \simeq g$ rel K or $f \simeq_K g$. This is an equivalence relation.

When $K = \{x_0\}$, one calls a homotopy of based maps relative to K simply as a *homotopy of based maps* or *based homotopy*. We write $f \simeq g$ if such a homotopy exists.

Definition 3.0.2 (Homotopy equivalence)**.** Let X and Y be topological spaces and let $K \subseteq X$.

1. A continuous map (Definition 1.3.1) $f : X \to Y$ is a *homotopy equivalence relative to K* if there exists a continuous map $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{id}_X \ \text{ rel } K \quad \text{and} \quad f \circ g \simeq \mathrm{id}_Y \ \text{ rel } f(K).$$

(Definition 3.0.1) That is, there exist homotopies (Definition 3.0.1) $H : X \times [0,1] \to X$ and $G : Y \times [0,1] \to Y$ satisfying

$$\begin{aligned} H(x,0) &= (g \circ f)(x), & H(x,1) &= x, & H(k,t) &= k \text{ for all } k \in K, t \in [0,1], \\ G(y,0) &= (f \circ g)(y), & G(y,1) &= y, & G(f(k),t) &= f(k) \text{ for all } k \in K, t \in [0,1]. \end{aligned}$$

In this case, the map g is called a *homotopy inverse relative to K of f*, and the pairs (X,K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K*, denoted $(X,K) \simeq (Y, f(K))$ rel K. This is an eqiuvalence relation.

A continuous map $f : X \to Y$ is a *homotopy eqiuvalence* if it is a homotopy equivalence relative to $\emptyset$. A *homotopy inverse of f* is then simply a homotopy inverse of f

relative to $\emptyset$. In this case, and the spaces X and Y are said to be *homotopy equivalent*, denoted $X \simeq Y$. This is an equivalence relation.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset with $x_0 \in K$.

   A continuous map $f : (X, x_0) \to (Y, y_0)$ of pointed spaces is a *homotopy equivalence relative to K* if there exists a continuous map $g : (Y, y_0) \to (X, x_0)$ such that

$$g \circ f \simeq \mathrm{id}_X \ \text{ rel } K \quad \text{and} \quad f \circ g \simeq \mathrm{id}_Y \ \text{ rel } f(K),$$

   where the homotopies (Definition 3.0.1) are homotopies of based maps relative to K and $f(K)$, respectively.

   That is, there exist homotopies of based maps $H : X \times [0,1] \to X$ and $G : Y \times [0,1] \to Y$ satisfying

$$\begin{aligned} H(x,0) &= (g \circ f)(x), & H(x,1) &= x, & H(k,t) &= k \text{ for all } k \in K, t \in [0,1], \\ G(y,0) &= (f \circ g)(y), & G(y,1) &= y, & G(f(k),t) &= f(k) \text{ for all } k \in K, t \in [0,1], \end{aligned}$$

   with $H(x_0, t) = x_0$ and $G(y_0, t) = y_0$ for all t.

   The map g is called a *homotopy inverse relative to K of f*, and the pairs (X, K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K*, denoted $(X, K) \simeq (Y, f(K))$ rel K. This is an equivalence relation.

   A continuous map $f : (X, x_0) \to (Y, y_0)$ of pointed spaces is a *homotopy equivalence* if it is a homotopy equivalence relative to $\{x_0\}$. A *homotopy inverse of f* is then simply a homotopy inverse relative to $\{x_0\}$. In this case, the pointed spaces (X, x_0) and (Y, y_0) are said to be *homotopy equivalent*, denoted $(X, x_0) \simeq (Y, y_0)$.

   In each case, topological spaces that are homotopy equivalent are said to be of the same *homotopy type*.

Definition 3.0.3 (Homotopy class of maps relative to a subset)**.** Let X and Y be topological spaces (Definition 1.1.1) and let $K \subseteq X$. Let $C(X, Y)$ denote the set of all continuous maps (Definition 1.3.1) $f : X \to Y$.

1. Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy relative to K (Definition 3.0.1)

$$H : X \times [0,1] \to Y$$

   such that

$$H(x,0) = f(x), \quad H(x,1) = g(x),$$

   and

$$H(k,t) = f(k) = g(k) \quad \text{for all } k \in K, t \in [0,1].$$

   The *homotopy class of maps relative to K* containing a map $f : X \to Y$ is denoted by $[f]_K$.

   Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\emptyset$.

   The *homotopy class of maps* containing a map $f : X \to Y$ is denoted by $[f]$.

   The set of homotopy classes of maps may often be denoted by $[X, Y]$.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition 1.3.3) and let $K \subseteq X$ be a subset containing x_0. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \to Y$ with $f(x_0) = y_0$.

   Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy of based maps relative to K

   $$H : X \times [0, 1] \to Y$$

   such that for all $x \in X$,

   $$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

   and for all $k \in K$ and $t \in [0, 1]$,

   $$H(k, t) = f(k) = g(k),$$

   particularly ensuring the basepoint is fixed throughout,

   $$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

   In the case that $K = \{x_0\}$, one simply says that f and g are *homotopic (as based maps)*.

   The *homotopy class relative to K* containing $f : (X, x_0) \to (Y, y_0)$ is denoted by $[f]_K$.

   Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\{x_0\}$.

   The *homotopy class* containing a map $f : (X, x_0) \to (Y, y_0)$ is denoted by $[f]$.

   The set of homotopy classes of pointed maps $(X, x_0) \to (Y, y_0)$ may often be denoted by $[(X, x_0), (Y, y_0)]$ or by notations such as $[X, Y]_*$, $[X, Y]_\bullet$, or $[X, Y]$ if the base points are clear.

Definition 3.0.4. The *homotopy category of topological spaces*, denoted hTop, is the category whose objects are topological spaces (Definition 1.1.1) and whose morphisms are homotopy classes (Definition 3.0.3) of continuous maps (Definition 1.3.1). In other words, for objects X and Y, the set of morphisms is defined as $\mathrm{Hom}_{\mathrm{hTop}}(X, Y) = [X, Y] = C(X, Y)/\simeq$.

Proposition 3.0.5. Composition in the homotopy category of topological spaces (Definition 3.0.4) is well-defined. If $f_1, f_2 : X \to Y$ are homotopic and $g_1, g_2 : Y \to Z$ are homotopic, then the compositions $g_1 \circ f_1$ and $g_2 \circ f_2$ are homotopic as maps from X to Z. That is, $[g] \circ [f] = [g \circ f]$ is independent of the choice of representatives.

Theorem 3.0.6. There exists a canonical functor $Q : \mathrm{Top} \to \mathrm{hTop}$ which is the identity on objects and maps each continuous map f to its homotopy class $[f]$. This functor is full and essentially surjective.

Definition 3.0.7 (Path in a topological space)**.** Let X be a topological space (Definition 1.1.1), let $x_0, x_1 \in X$, and let $[0, 1] \subset \mathbb{R}$ be the closed unit interval (Definition 1.5.5) equipped with the subspace topology (Definition 1.1.26) from $\mathbb{R}$.

1. A *(continuous) path in X from x_0 to x_1* is a continuous map (Definition 1.3.1) $\gamma : [0, 1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. More generally, a *path in X* is a path from some point $x_0 \in X$ to some point $x_1 \in X$ in this sense.

2. A *loop in X based at $x_0 \in X$* or a *closed path in X based at $x_0 \in X$* is a path $\gamma : [0,1] \to X$ in X from x_0 to x_0, that is, a continuous map γ such that $\gamma(0) = \gamma(1) = x_0$. When a basepoint x_0 is fixed and understood from the context, a *loop in X* means a loop in X based at this chosen basepoint.

Definition 3.0.8 (Homotopy groups of a pointed topological space)**.** For any pointed topological space (Definition 1.3.3) (X, x_0) and integer $n \geq 0$, the *n-th homotopy set of X at x_0*, denoted $\pi_n(X, x_0)$ or simply $\pi_n(X)$ when the base point x_0 is understood or a choice of base point does not really matter, is defined as the set of all homotopy classes (rel. ∂I^n) (Definition 3.0.3) of paired maps

$$f : (I^n, \partial I^n) \to (X, \{x_0\}),$$

where $I^n = [0,1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary (Definition 1.2.18) thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed maps (Definition 1.3.3)

$$f : (S^n, s_0) \to (X, x_0),$$

where S^n is the n-sphere (Definition 1.5.4) and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$ (Definition 3.0.3).

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *nth homotopy group of (X, x_0)*.

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops (Definition 3.0.7) $\gamma : [0,1] \to X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\mathrm{Top}_* \to \mathbf{Sets}$,
2. For $n = 1$, it is a functor $\mathrm{Top}_* \to \mathbf{Grp}$,
3. For $n \geq 2$, it is a functor $\mathrm{Top}_* \to \mathbf{Ab}()$.

Given a map $f : (X, x_0) \to (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \to \pi_n(Y, y_0)$ is often denoted by f_*.

The homotopy set $\pi_n(X, x_0)$ is naturally isomorphic to the relative homotopy set $\pi_n(X, \{x_0\}, x_0)$ (Definition 3.0.9).

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\mathrm{Map}_*((S^n, s_0), (X, x_0)))$.

Definition 3.0.9 (Relative homotopy groups of pointed pairs)**.** For any pointed pair of topological spaces (X, A, x_0) and an integer $n \geq 1$, the *n-th relative homotopy set of (X, A) at x_0*, denoted $\pi_n(X, A, x_0)$, is defined as the set of all homotopy classes (rel. J^{n-1}) (Definition 3.0.3) of maps of triples

$$f : (I^n, \partial I^n, J^{n-1}) \to (X, A, \{x_0\}),$$

where $I^n = [0,1]^n$ is the n-cube, ∂I^n is its boundary (Definition 1.2.18), and

$$J^{n-1} := \overline{\partial I^n \setminus (I^{n-1} \times \{1\})} = (I^{n-1} \times \{0\}) \cup (\partial I^{n-1} \times I) \subseteq \partial I^n.$$

Explicitly, an element of $\pi_n(X, A, x_0)$ is represented by a continuous map (Definition 1.3.1) $f : I^n \to X$ such that:

1. $f(\partial I^n) \subseteq A$,
2. $f(J^{n-1}) = \{x_0\}$.

Equivalently, $\pi_n(X, A, x_0)$ can be defined as the set of all homotopy classes of maps of pairs

$$f : (D^n, S^{n-1}, s_0) \to (X, A, x_0),$$

where D^n is the n-disk, S^{n-1} is its boundary sphere, and $s_0 \in S^{n-1}$ is a base point.

For $n \geq 2$, $\pi_n(X, A, x_0)$ is a group under the concatenation of maps in the first coordinate, and for $n \geq 3$, it is abelian. As such:

1. For $n = 1$, the assignment $(X, A, x_0) \mapsto \pi_1(X, A, x_0)$ is a functor $\mathbf{Top}_{*,\mathrm{pair}} \to \mathbf{Sets}$,
2. For $n = 2$, it is a functor $\mathbf{Top}_{*,\mathrm{pair}} \to \mathbf{Grp}$,
3. For $n \geq 3$, it is a functor $\mathbf{Top}_{*,\mathrm{pair}} \to \mathbf{Ab}$.

For $n \geq 2$, $\pi_n(X, A, x_0)$ is called the *n-th relative homotopy group of (X, A) at x_0*. Given a map $f : (X, A, x_0) \to (Y, B, y_0)$ of pointed pairs, the induced map $\pi_n(X, A, x_0) \to \pi_n(Y, B, y_0)$ is denoted by f_*.

Definition 3.0.10 (Weak homotopy equivalence)**.** Let $f : X \to Y$ be a continuous map (Definition 1.3.1) between topological spaces. The map f is a *weak homotopy equivalence* if, for every choice of basepoint $x_0 \in X$ with $y_0 = f(x_0)$, the induced maps on homotopy groups (Definition 3.0.8)

$$f_* : \pi_n(X, x_0) \to \pi_n(Y, y_0)$$

are isomorphisms (Definition C.0.2) for all integers $n \geq 0$.

4. Singular homology and cohomology

Definition 4.0.1. Let V be a real vector space of finite dimension. A *k-simplex in topology* (or a *geometric k-simplex*) is the convex hull of $k+1$ affinely independent points $v_0, v_1, \ldots, v_k \in V$, and is denoted by

$$[v_0, v_1, \ldots, v_k] := \left\{ \sum_{i=0}^{k} t_i v_i \;\middle|\; t_i \geq 0,\ \sum_{i=0}^{k} t_i = 1 \right\}.$$

It is also standard to talk of the *standard topological n-simplex* — the topological space, often denoted by notations such as Δ^n or $|\Delta^n|$ defined as the subset of Euclidean space $\mathbb{R}^{n+1}$ given by

$$|\Delta^n| = \left\{(t_0, t_1, \ldots, t_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^{n} t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i\right\}$$

equipped with the induced topology from the usual Euclidean topology on $\mathbb{R}^{n+1}$.

(♠ TODO: comment on how $|\Delta^n|$ makes sense via a geometric realization)

Definition 4.0.2 (Singular simplices)**.** Let X be a topological space (Definition 1.1.1). For each integer $n \geq 0$, A *singular n-simplex in X* is a continuous map (Definition 1.3.1)

$$\sigma : \Delta^n \to X.$$

where Δ^n is the standard topological n-simplex (Definition 4.0.1). The set of all singular n-simplices in X is denoted by $S_n(X)$.

Definition 4.0.3 (Singular chain group with coefficients)**.** Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.26), and let $S_n(X)$ be the set of singular n-simplices in X (Definition 4.0.2).

1. Let R be a ring.
   (a) The *singular n-chain group of X with coefficients in R* is the free R-module $C_n(X; R)$ whose elements are finite formal linear combinations

   $$\sum_i r_i \, \sigma_i, \quad \text{with } \sigma_i \in S_n(X), \ r_i \in R.$$

   Elements of $C_n(X; R)$ are called *singular n-chains in X with coefficients in R*.
   (b) If $A \subseteq X$ is a subspace, the quotient groups

   $$C_n(X, A; R) = C_n(X; R)/C_n(A; R)$$

   may be referred to as the *relative singular n-chain groups of X in A with coefficients in R* and elements of this group may be referred to as *relative singular n-chains in X relative to A with coefficients in R*.

   In either case, when $R = \mathbb{Z}$, the ring R may be suppressed from notation, so we may write $C_n(X)$ and $C_n(X, A)$ for $C_n(X, \mathbb{Z})$ and $C_n(X, A, \mathbb{Z})$ respectively.
2. Let G be an abelian group (Definition C.0.1).
   (a) More generally, the *singular n-chain group of X with coefficients in G* is the abelian group

   $$C_n(X; G) = C_n(X; \mathbb{Z}) \otimes_{\mathbb{Z}} G.$$

   (**??**) In particular, an element of $C_n(X; G)$ is a finite formal sum

   $$\sum_i g_i \sigma_i, \quad g_i \in G, \sigma_i \in S_n(X).$$

   Elements of $C_n(X; G)$ are called *singular n-chains in X with coefficients in G*.
   (b) Similarly, the *relative singular n-chain group of X in A with coefficients in G* is the quotient group

   $$C_n(X, A; G) = C_n(X; G)/C_n(A; G)$$

Elements of $C_n(X, A; G)$ are called *singular n-chains in X relative to A with coefficients in G*.

When $G = R$ is a ring, note that the notions of $C_n(X; R)$ where R is taken as a ring and where R is taken as an abelian group are equivalent; similarly for $C_n(X, A; R)$.

We note that, in general $C_n(X; G) \cong C_n(X, \emptyset; G)$.

Definition 4.0.4 (Singular chain complex with coefficients)**.** Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.26), and let G be an abelian group.

For each $n \geq 1$, define the *boundary operator*

$$\partial_n : C_n(X; G) \to C_{n-1}(X; G)$$

(Definition 4.0.3) by

$$\partial_n(\sigma) = \sum_{i=0}^{n} (-1)^i \, \sigma \circ \delta_i,$$

where $\delta_i : \Delta^{n-1} \to \Delta^n$ is the i-th face inclusion, and extend ∂_n to $C_n(X; G)$ linearly. For $n \leq 0$, we simply let $\partial_n = 0$

One can verify that $(C_n(X; G), \partial_n)$ forms a chain complex (Definition E.0.1) in the abelian category of abelian groups. This chain complex is called the *(absolute) singular chain complex of X with coefficients in G* and is often denoted by $C_\bullet(X; G)$.

Then the boundary maps above induce maps

$$C_n(X, A; G) \to C_{n-1}(X, A; G)$$

on the relative chain groups $C_n(X, A; G)$, yielding a chain complex of abelian groups whose boundary maps, often also denoted by ∂_n, are induced by the boundary operators ∂_n above; this chain complex may be called the *relative singular chain complex of the pair (X, A) with coefficients in G* and is often denoted by $C_\bullet(X, A; G)$.

In the case that $G = R$ is a (not necessarily commutative) ring, then the boundary maps of $C_\bullet(X; R)$ and $C_\bullet(X, A; R)$ are left R-module homomorphisms:

$$\partial \left(\sum_i r_i \sigma_i \right) = \sum_i r_i \partial(\sigma_i)$$

Definition 4.0.5 (Singular homology with coefficients)**.** Let X be a topological space (Definition 1.1.1), let $A \subseteq X$ be a subspace (Definition 1.1.26), and let G be an abelian group.

The *(absolute) n-th singular homology group of X with coefficients in G* is the homology group (Definition E.0.2)

$$H_n(X; G) = H_n(C_*(X; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X; G)$ is the singular chain complex of X with coefficients in G and $\partial_n : C_n(X; G) \to C_{n-1}(X; G)$ are the boundary maps.

Given a subspace (Definition 1.1.26) $A \subseteq X$, the *n-th relative singular homology group of (X, A) with coefficients in G* is defined as

$$H_n(X, A; G) = H_n(C_*(X, A; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X, A; G)$ is the relative singular chain complex of (X, A) with coefficients in G and $\partial_n : C_n(X, A; G) \to C_{n-1}(X, A; G)$ are the boundary maps.

We may denote $H_n(X; \mathbb{Z})$ and $H_n(X, A; \mathbb{Z})$ by $H_n(X)$ and $H_n(X, A)$ respectively.

In the case that $G = R$ is a ring, $H_n(X; R)$ and $H_n(X, A; R)$ have the structure of left R-modules (Definition D.0.2) since the corresponding chain complexes are chain complexes of left R-modules..

Definition 4.0.6 (Singular cochain complex of a topological space)**.** Let X be a topological space (Definition 1.1.1) and R a commutative ring (Definition D.0.1) with 1. The *singular n-cochain group of X with coefficients in R* is the R-module

$$C^n(X; R) = \operatorname{Hom}_R(C_n(X; R), R) .$$

(Definition D.0.3) (Definition 4.0.3) We define *coboundary operators*

$$\delta^n : C^n(X; R) \to C^{n+1}(X; R), \quad \delta^n(\varphi) = \varphi \circ \partial_{n+1}.$$

The pair $(C^n(X; R), \delta^n)_n$ form a cochain complex (Definition E.0.1) called the *singular cochain complex of X with coefficients in R*. Given a subspace $A \subseteq X$, the *relative singular n-cochain group of (X, A) with coefficients in R* is the R-module

$$C^n(X, A; R) = \operatorname{Hom}_R(C_n(X, A; R), R) .$$

The coboundary operators δ^n induce maps $C^n(X, A; R) \to C^{n+1}(X, A; R)$ for which the $C^n(X, A; R)$ form a cochain complex, called the *relative singular cochain complex of (X, A) with coefficients in R*

Definition 4.0.7. Let X be a topological space (Definition 1.1.1) and R a commutative ring (Definition D.0.1) with 1. The *n-th singular cohomology group of X with coefficients in R* is

$$H^n(X; R) = \ker(\delta^n) / \operatorname{im}(\delta^{n-1}).$$

For a subspace $A \subseteq X$ the *relative cohomology groups of (X, A)* are

$$H^n(X, A; R) = H^n(C^*(X, A; R)).$$

In other words, these singular cohomology groups are the cohomology R-modules (Definition E.0.2) of the singular cochain complex $C^*(X; R)$ and the relative singular cochain complex $C^*(X, A; R)$ (Definition 4.0.6) respectively.

5. CW complexes

Definition 5.0.1 (CW Complex)**.** A **CW complex** is a cellular complex X that satisfies the following conditions:

- **Hausdorff:** X is a Hausdorff space (Definition 1.1.25).
- **C (Closure Finiteness):** The closure of each cell $\overline{e^n_\alpha}$ is contained in a finite union of cells of dimension less than or equal to n.
- **W (Weak Topology):** A subset $A \subseteq X$ is closed if and only if $A \cap \overline{e^n_\alpha}$ is closed for every cell e^n_α.

6. Fibrations

6.1. Hurewicz fibrations.

Definition 6.1.1 (Homotopy Lifting Property)**.** Let $p : E \to B$ be a continuous map. The map p is said to have the *homotopy lifting property (HLP) with respect to a topological space X* if, for every homotopy (Definition 3.0.1) $H : X \times I \to B$ and every map $f : X \to E$ such that $p \circ f = H(\cdot, 0)$, there exists a homotopy $\tilde{H} : X \times I \to E$ such that:

1. $\tilde{H}(\cdot, 0) = f$ (the lift starts at the given map);
2. $p \circ \tilde{H} = H$ (the diagram commutes).

In terms of diagrams, p satisfies the HLP with respect to X if every commutative solid square of the following form admits a dashed filler $\tilde{H}$:

$$\begin{tikzcd} X \times \{0\} \arrow[r, "f"] \arrow[d, hook] & E \arrow[d, "p"] \\ X \times I \arrow[r, "H"'] \arrow[ur, dashed, "\exists \tilde{H}"] & B \end{tikzcd}$$

Definition 6.1.2. Let $p : E \to B$ be a continuous map of topological spaces (Definition 1.1.1). The map p is called a *Hurewicz fibration* (or simply a *fibration*) if it has the homotopy lifting property (Definition 6.1.1) with respect to all topological spaces X.

Equivalently, for every topological space X and every commutative diagram of the form:

$$\begin{tikzcd} X \times \{0\} \arrow[r, "f"] \arrow[d, "i"'] & E \arrow[d, "p"] \\ X \times I \arrow[r, "H"] \arrow[ur, dashed, "\exists \tilde{H}"] & B \end{tikzcd}$$

where $I = [0, 1]$ is the unit interval and $i(x) = (x, 0)$, there exists a continuous map $\tilde{H} : X \times I \to E$ such that $p \circ \tilde{H} = H$ and $\tilde{H} \circ i = f$.

6.2. Serre fibrations.

Definition 6.2.1 (Serre Fibration)**.** A continuous map (Definition 1.3.1) $p : E \to B$ of topological spaces (Definition 1.1.1) is called a *Serre fibration* (or a *weak fibration*) if it has the homotopy lifting property (Definition 6.1.1) with respect to all CW complexes (Definition 5.0.1).

Equivalently, for every $n \geq 0$, and every commutative diagram of the form:

$$\begin{tikzcd} I^n \times \{0\} \arrow[r, "f"] \arrow[d, "i"'] & E \arrow[d, "p"] \\ I^n \times I \arrow[r, "H"] \arrow[ur, dashed, "\exists \tilde{H}"] & B \end{tikzcd}$$

where $I = [0, 1]$ is the unit interval (Definition 1.5.5) and I^n is the n-dimensional cube (Definition 1.5.6), there exists a continuous map $\tilde{H} : I^n \times I \to E$ making the entire diagram commute.

Lemma 6.2.2. Any Hurewicz fibration (Definition 6.1.2) is a Serre fibration (Definition 6.2.1).

Proof. This holds trivially from the definitions — Hurewicz fibrations satisfy HLP (Definition 6.1.1) for all topological spaces whereas Serre fibrations satisfy HLP for all disks D^n. □

6.3. **Fiber bundles.**

Definition 6.3.1. Let F be a topological space (Definition 1.1.1). A continuous map (Definition 1.3.1) $\pi : E \to B$ of topological spaces (Definition 1.1.1) is called a *fiber bundle with fiber F* if for every point $b \in B$, there exists an open neighborhood (Definition 1.1.16) $U \subseteq B$ containing b and a homeomorphism (Definition 1.3.2) $\phi : \pi^{-1}(U) \to U \times F$ such that the following diagram commutes:

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & U \times F \arrow[dl, "pr_1"] \\ & U & \end{tikzcd}$$

where $pr_1 : U \times F \to U$ is the projection onto the first factor. The space E is called the *total space*, B is the *base space*, and the homeomorphism ϕ is called a *local trivialization of the fiber bundle over U*.

It is also reasonable to write $F \hookrightarrow E \twoheadrightarrow B$ or simply $F \to E \to B$ to express the existence of a fiber bundle.

Definition 6.3.2. A topological space is *paracompact* if it is Hausdorff (Definition 1.1.25) and every open cover of X admits an open refinement that is locally finite.

See Theorem 6.3.3, which posits that nice enough topological spaces are paracompact.

Theorem 6.3.3. The following are paracompact (Definition 6.3.2) topological spaces:

1. Metric spaces (Definition 1.7.1) (Stone's theorem)
2. Topological manifolds
3. Compact (Definition 1.2.14) Hausdorff (Definition 1.1.25) spaces
4. CW-complexes (Definition 5.0.1)

Proposition 6.3.4. Let $\pi : E \to B$ be a fiber bundle (Definition 6.3.1) with fiber F. If B is a paracompact space (Definition 6.3.2), then π is a Hurewicz fibration (Definition 6.1.2).

Proposition 6.3.5. Let $\pi : E \to B$ be a fiber bundle (Definition 6.3.1). Then π is a Serre fibration (Definition 6.2.1).

Appendix A. Set theory

Definition A.0.1. Let A be a set.

- The set A is said to be *countably infinite* (or simply *countable*) if there exists a bijection $f : \mathbb{N} \to A$.
- The set A is said to be *finite* if there exists some $n \in \mathbb{N}$ and a bijection $g : \{1, 2, \ldots, n\} \to A$.
- The set A is said to be *at most countable* if it is either finite or countably infinite.
- The set A is said to be *uncountable* if it is not at most countable.

Definition A.0.2. Let X and Y be sets and let $f : X \to Y$ be a function.

- The function f is said to be *injective* (or *one-to-one*) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- The function f is said to be *surjective* (or *onto*) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is *bijective* if it is both injective and surjective. In this case, there exists a unique *inverse map* $f^{-1} : Y \to X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

Appendix B. Category Theory

Definition B.0.1 (Category)**.** A *category* $\mathcal{C}$ consists of the following data:

- A class of *objects* denoted $\mathrm{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, a class

$$\mathrm{Hom}_{\mathcal{C}}(X, Y)$$

  of *morphisms* (also called *arrows* or *homs*). If the category $\mathcal{C}$ is clear, then this *hom-class* is also denoted by $\mathrm{Hom}(X, Y)$. It may also be denoted by $\mathrm{hom}_{\mathcal{C}}(X, Y)$ or $\mathrm{hom}(X, Y)$, especially to distinguish from other types of hom's (e.g. internal hom's)
- For each triple of objects X, Y, Z, a composition law

$$\circ : \mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Z),$$

  denoted $(g, f) \mapsto g \circ f$.
- For each object X, an *identity morphism*

$$\mathrm{id}_X \in \mathrm{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \mathrm{Hom}_{\mathcal{C}}(X,Y)$, $g \in \mathrm{Hom}_{\mathcal{C}}(Y,Z)$, and $h \in \mathrm{Hom}_{\mathcal{C}}(Z,W)$,
$$h \circ (g \circ f) = (h \circ g) \circ f.$$
- (Identity) For all $f \in \mathrm{Hom}_{\mathcal{C}}(X,Y)$,
$$\mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X .$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \mathrm{Ob}(\mathcal{C})$, i.e. to denote that X is an object of of $\mathcal{C}$.

We may call a category as above an *ordinary category* to distinguish this notion from the notions of *categories enriched in monoidal categories* or higher/n-categories. (♠ TODO: TODO: define n-categories)

A category as defined above may be called called a *large category* or a *class category* to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a *category* may often refer to a locally small category (Definition B.0.2), which is a category whose hom-classes are all sets.

Definition B.0.2 (Locally small category)**.** A (large) category (Definition B.0.1) $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, the collection $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ of morphisms between them is a (small) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\mathrm{Ob}(\mathcal{C})$ of objects is a set.

Remark B.0.3. Many "concrete" categories considered in "classical mathematics" or outside of more "abstract" category theory tend to be locally small. For example, the categories of sets, groups, R-modules, vector spaces, topological spaces, schemes, manifolds, sheaves on "small enough" sites are all locally small.

Appendix C. Group Theory

Definition C.0.1 (Groups)**.** A *group* is a monoid such that every element has an inverse.

In other words, a group is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \to G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has
$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$
2. (Identity element) There exists an identity element, i.e. an element $e \in G$ such that for all $g \in G$,
$$e \cdot g = g \cdot e = g.$$

3. (Inverse elements) For all $g \in G$, there exists an inverse element of g, i.e. an element $g^{-1} \in G$ such that
$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

A group $(G, \cdot)$ is often simply written as G, when the notation for the binary operation $\cdot$ is clear.

Assuming that the notation for the identity element of a group is not explicitly specified,

1. if the group operation is written multiplicatively, then one usually implicitly writes e or 1 as the identity of G, and
2. If the group operation is instead written additively, then one usually implicitly write 0 as the identity of G.

An *abelian group* or synonymously, a *commutative group*, is a group $(G, \cdot)$ whose binary operation $\cdot$ is *abelian* or *commutative*, i.e. satisfies
$$g \cdot h = h \cdot g$$
for all $g, h \in G$. A group operation written additively, i.e. with $+$, is almost usually abelian. An abelian group's group operation may still often be written multiplicatively.

An abelian group is equivalent to a $\mathbb{Z}$-module.

Definition C.0.2 (Group homomorphism)**.** Let $(G, \cdot)$ and $(H, *)$ be groups (Definition C.0.1). A map $f : G \to H$ is called a *group homomorphism* if for all $g_1, g_2 \in G$ one has
$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small (Definition B.0.2) category (Definition B.0.1), called the *category of groups*.

If f is bijective (Definition A.0.2), then f is called a *group isomorphism*. .

Appendix D. Ring Theory

Definition D.0.1. A *commutative (unital) ring* is a ring $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. "commutative" algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition D.0.2. Let R be a not-necessarily commutative ring.

1. A *left R-module* is an abelian group $(M, +)$ together with an operation $R \times M \to M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
   - $r(m + n) = rm + rn$,
   - $(r + s)m = rm + sm$,
   - $(rs)m = r(sm)$,
   - $1_R m = m$ where 1_R is the multiplicative identity of R.

2. A *right R-module* is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \to M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
   - $(m + n)r = mr + nr$,
   - $m(r + s) = mr + ms$,
   - $m(rs) = (mr)s$,
   - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings.
   An *R-S-bimodule* (or an *R-S-module* or an (R, S)-module, etc.) is an abelian group (Definition C.0.1) $(M, +)$ equipped with
   (a) a left action of R:
   $$R \times M \to M, \quad (r, m) \mapsto r \cdot m,$$
   making M a left R-module (Definition D.0.2),
   (b) a right action of S:
   $$M \times S \to M, \quad (m, s) \mapsto m \cdot s,$$
   making M a right S-module,
   such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,
   $$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$
4. A *two-sided R-module* (or *R-bimodule*) is an R-R-bimodule.

If R is a commutative ring (Definition D.0.1), then a left/right R-module can automatically be regarded as a two-sided R-module. As such, we simply talk about *R-modules* in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$-module. Moreover, any left R-module is equivalent to an $R - \mathbb{Z}$-bimodule (Definition D.0.2) and any right R-module is equivalent to an $\mathbb{Z} - R$-bimodule (Definition D.0.2). Given a left/right/two-sided R-module, its *natural bimodule structure* will refer to its structure as a R-$\mathbb{Z}$/$\mathbb{Z}$-R/R-R bimodule. In this way, many definitions associated with the notions of left/right/two-sided R-modules can be defined as special cases for definitions for R-S-bimodules.

Definition D.0.3 (Hom of left/right/bi-modules)**.** Let R, S, T be (not necessarily commutative) rings.

1. Let M and N be left R-modules (Definition D.0.2). The *homomorphism group of left R-modules from M to N* is the abelian group

$$\mathrm{Hom}(M, N) = \mathrm{Hom}_R(M, N) = \mathrm{Hom}_{R-\mathbb{Z}}(M, N) := \{f : M \to N \mid f \text{ is a left } R\text{-module homomorphism}\}.$$

2. Let M and N be right R-modules (Definition D.0.2). The *homomorphism group of right R-modules from M to N* is the abelian group

$$\mathrm{Hom}(M, N) = \mathrm{Hom}_R(M, N) = \mathrm{Hom}_{\mathbb{Z}-R}(M, N) := \{f : M \to N \mid f \text{ is a right } R\text{-module homomorphism}\}.$$

3. Let S be a (not necessarily commutative ring) and let M and N be $R - S$-bimodules (Definition D.0.2). The *homomorphism group of R-S-bimodules from M to N* is the

abelian group

$$\mathrm{Hom}(M,N) = \mathrm{Hom}_{R-S}(M,N) \coloneqq \{f : M \to N | f \text{ is a } R-S\text{-bimodule homomorphism}\}$$

In each case, $\mathrm{Hom}(M,N)$ has a natural structure of an *abelian group* given by *pointwise addition*: for $f, g \in \mathrm{Hom}(M,N)$,

$$(f+g)(m) := f(m) + g(m),$$

and the zero morphism 0 given by $0(m) := 0_N$ acts as the identity element. The additive inverse $-f$ is defined by $(-f)(m) := -f(m)$. Moreover, depending on bi-module structures that M and N may be carrying, $\mathrm{Hom}(M,N)$ may itself carry additional module structures:

- In case that M is a $R-S$-bimodule and N is a $R-T$-bimodule, $\mathrm{Hom}_R(M,N)$, the group of left R-module homomorphisms, is an $S-T$-bimodule as follows:
$$(s \cdot f \cdot t)(m) = f(m \cdot s) \cdot t \quad f \in \mathrm{Hom}_R(M,N), s \in S, t \in T.$$
- Dually, in case that M is a $S-R$-bimodule and N is a $T-R$-bimodule, $\mathrm{Hom}_R(M,N)$, the group of right R-module homomorphisms, is an $S-T$-bimodule as follows:
$$(s \cdot f \cdot t)(m) = f(s \cdot m) \cdot t \quad f \in \mathrm{Hom}_R(M,N), s \in S, t \in T.$$

Some cases of interest may be when R, S, or T is in fact $\mathbb{Z}$ — these allow us to see module structures on $\mathrm{Hom}(M,N)$ even when M and N are one-sided modules.

(♠ TODO: state this as a theorem) We furthermore note that $\mathrm{Hom}_R(-,-)$ yields biadditive functors

$$\mathrm{Hom}_R(-,-) : {}_R\mathbf{Mod}_S^{\mathrm{op}} \times {}_R\mathbf{Mod}_T \to {}_S\mathbf{Mod}_T$$
$$\mathrm{Hom}_R(-,-) : {}_S\mathbf{Mod}_R^{\mathrm{op}} \times {}_T\mathbf{Mod}_R \to {}_S\mathbf{Mod}_T.$$

Appendix E. Homological algebra

Definition E.0.1 (Chain complex in a preadditive category)**.** Let $\mathcal{A}$ be a preadditive category and let I be a totally ordered set (typically $\mathbb{Z}$, but $I \subseteq \mathbb{Z}$ is also allowed).

1. A *chain complex* (or *homological (chain) complex*) $(K_\bullet, d_\bullet)$ in $\mathcal{A}$ indexed by I is the homological convention for sequences with decreasing degrees. It consists of:
   - Objects $\{K_i\}_{i\in I}$ in $\mathcal{A}$, called the *terms in degree i*,
   - Morphisms $d_i : K_i \to K_{i-1}$ in $\mathcal{A}$, called the *boundary maps* or *differentials in degree i*,

   such that for every $i \in I$, $d_{i-1} \circ d_i = 0$. That is,
$$K_\bullet : \cdots \xrightarrow{d_{i+1}} K_i \xrightarrow{d_i} K_{i-1} \xrightarrow{d_{i-1}} K_{i-2} \to \cdots$$
   with $d_{i-1}d_i = 0$ for all i. We typically use the notation $K_\bullet = (K_i, d_i)_{i\in I}$.
2. Dually, a *cochain complex* (or *cohomological complex*) $(K^\bullet, d^\bullet)$ in $\mathcal{A}$ follows the cohomological convention with increasing degrees. It consists of objects $\{K^i\}_{i\in I}$ and *coboundary maps* $d^i : K^i \to K^{i+1}$ such that $d^{i+1} \circ d^i = 0$:
$$K^\bullet : \cdots \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \xrightarrow{d^{i+1}} K^{i+2} \to \cdots$$

We typically use the notation $K^\bullet = (K^i, d^i)_{i\in I}$.

3. Let $K_\bullet = (K_i, d_i^K)$ and $L_\bullet = (L_i, d_i^L)$ be chain complexes (Definition E.0.1) in $\mathcal{A}$ indexed by the same set I. A *morphism of chain complexes* (or *chain map*)

$$f_\bullet : K_\bullet \to L_\bullet$$

consists of morphisms $f_i : K_i \to L_i$ for all $i \in I$, such that for every $i \in I$, the following diagram commutes:

$$\begin{array}{ccc} K_i & \xrightarrow{d_i^K} & K_{i-1} \\ \downarrow f_i & & \downarrow f_{i-1} \\ L_i & \xrightarrow{d_i^L} & L_{i-1} \end{array}$$

i.e., $d_i^L \circ f_i = f_{i-1} \circ d_i^K$.

A *morphism of cochain complexes* $f^\bullet : K^\bullet \to L^\bullet$ is defined similarly, satisfying the commutativity condition $d_L^i \circ f^i = f^{i+1} \circ d_K^i$.

The collection of these objects and morphisms forms a category. Notation for these categories is as follows:

- $\mathrm{Ch}(\mathcal{A})$ or $\mathbf{Ch}(\mathcal{A})$ is often used as a general term.
- To be explicit about the indexing convention, one uses $\mathrm{Ch}_\bullet(\mathcal{A})$ for chain complexes and $\mathrm{Ch}^\bullet(\mathcal{A})$ (or sometimes $\mathrm{CoCh}(\mathcal{A})$) for cochain complexes.
- The set of chain maps between two complexes is denoted by $\mathrm{Hom}_{\mathrm{Ch}(\mathcal{A})}(K_\bullet, L_\bullet)$; it is an abelian group under pointwise addition $(f+g)_i = f_i + g_i$.

Notations used to write chain complexes are *homological conventions* whereas notations used to write cochain complexes are *cohomological conventions*. In homological conventions, indices are written as subscripts and decrease when "following the differential maps". In cohomological conventions, indices are written as superscripts and incrase when "following the diferential maps". The theories of chain complexes and of cochain complexes are not genuinely different, however, as the conventions are essentially notational — starting from a chain complex $(K_\bullet, d_\bullet)$, one can obtain a cochain complex $(K^\bullet, d^\bullet)$ simply by defining $K^i = K_{-i}$ and d^i as d_{-i} and vice versa. As such, cochain complexes may be simply called *chain complexes* when careful distinctions between the conventions are unnecessary. Homological and cohomological notational conventions persist throughout homological algebra, see e.g. spectral sequences, double complexes, etc, but again the distinction between such conventions are merely notational.

Definition E.0.2 (Chain complexes and their (co)homology objects)**.** Let $\mathcal{A}$ be an abelian category.

- For a cochain complex $K^\bullet$, its *cohomology object in degree i* is defined as the quotient of the object of i-cocycles by the object of i-coboundaries:

$$H^i(K^\bullet) := Z^i(K)/B^i(K) = \ker(d^i)/\operatorname{im}(d^{i-1}).$$

In fact, each H^i is a functor $H^i : \mathbf{Ch}^\bullet(\mathcal{A}) \to \mathcal{A}$ (Definition E.0.1); given a morphism $f^\bullet : K^\bullet \to L^\bullet$ of cochain complexes, it is thus customary to write $H^i(f^\bullet) : H^i(K^\bullet) \to H^i(L^\bullet)$ for the induced morphism on the ith cohomology objects.

- For a chain complex (Definition E.0.1) $K_\bullet$, its *homology object in degree i* is defined as the quotient of the object of i-cycles by the object of i-boundaries:

$$H_i(K_\bullet) := Z_i(K)/B_i(K). = \ker(d_i)/\operatorname{im}(d_{i+1}).$$

In fact, each H_i is a functor $H_i : \mathbf{Ch}_\bullet(\mathcal{A}) \to \mathcal{A}$ (Definition E.0.1); given a morphism $f_\bullet : K_\bullet \to L_\bullet$ of cochain complexes, it is thus customary to write $H_i(f_\bullet) : H_i(K_\bullet) \to H_i(L_\bullet)$ for the induced morphism on the ith homology objects.

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